ACADEMIC REGULATIONS & SYLLABUS

Faculty of Computer Science & Applications

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

B.Sc.(IT) Programme

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Computer Science & Applications

ACADEMIC REGULATIONS

B.Sc.(IT) Programme

(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)

CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in

www.charusat.ac.in

Year – 2019-2020

CHARUSAT

FACULTY OF COMPUTER SCIENCE AND APPLICATIONS

ACADEMIC RULES

Bachelor of Science in Information Technology (BSc.(IT) Programme

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Undergraduate programme Bachelor of Science in Information Technology

Minimum	6 semesters (3 academic years)
Maximum	10 semesters (5 academic years)

3. Eligibility & Mode of admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

4. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

5. Attendance

5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions.

However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

5.2 Student attendance in a course should be 80%.

6. Course Evaluation

6.1The performance of every student in each course will be evaluated as follows:

6.1.1Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

6.1.2Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

6.2 University Examination

6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

6.3Performance at Internal & University Examination will be done on the relative grading system.

7. Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

Grading Scheme

Range of Marks (%)	≥80	≥75 <80	≥70 <75	≥65 <70	≥60 <65	≥55 <60	≥50 <55	<50
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

SGPA = $\Sigma C_iG_i / \Sigma C_i$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and i = 1 to n, n = number of courses in the

semester

CGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

8. Detention Rule

8.1 No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.

8.2 A Student will not be allowed to move to third year if he/she has not cleared all the courses of first year.

8.3 A student will not be allowed to move to fourth year if he/she has not cleared all the courses of first and second year.

9. Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He should have cleared all evaluation components in every course; and

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

10 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA ≥7.5 & ≤10
First class:	CGPA ≥ 6.0 & <7.5
Second Class:	CGPA ≥ 5.0 & <6.0
Pass Class:	CGPA <5.0

11 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

FACULTY OF COMPUTER SCIENCE & APPLICATIONS

CHOICE BASED CREDIT SYSTEM

FOR

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

CHOICE BASED CREDIT SYSTEM

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses:

Foundation Courses, Core Courses and Elective Courses.

Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

Vision

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

Program Educational Objectives (PEOs)

Programme Educational Objectives (PEOs):- To prepare students to apply their knowledge and skills to be employed in IT professional careers and/or to continue their education in IT and/or related post graduate programmes.

PEO1 To provide students with core competence in computing, mathematical, statistical and management fundamentals necessary to formulate, analyze and solve information technology related problems and/or also to pursue advanced study or research.

PEO2 To provide students the concepts to analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.

PEO3 To assist students in the creation of an effective project plan.

PEO4 To provide specialization in emerging technologies such as software designing, open source technologies, data science, scripting technologies, embedded and IoT related applications.

PEO5 To Functions effectively in team and as an individual to accomplish desired goals.

PEO6 To Inculcate to maintain high professionalism and ethical standards, effective oral and written communication skills.

Program Outcomes (POs)

Programme Objectives(Pos):- Upon successful completion of the programme, graduate is expected to:

- PO1 Construct a sound knowledge, principles and skills in information technology.
- PO2 Develop the skills of analyzing, designing, implementing and managing information technology related solutions.
- PO3 Develop skills to integrate various technology solutions.
- PO4 Understand a broad business and real world perspective.
- PO5 Acquire specialization in emerging Information technologies.
- PO6 Build communication, team, leadership and interpersonal skills including technical documentation preparation of software systems.
- PO7 Gain Knowledge of contemporary issues in the social sciences and the humanities.
- PO8 An ability to recognize the importance of professional development by pursuing postgraduate studies or face competitive examinations that offer challenging and rewarding careers in computing.

TEACHING SCHEME

FOR

BSc.(IT) PROGRAMME

(1ST, 2ND & 3RD YEAR)

EFFECTIVE FROM

ACADEMIC YEAR 2019-20

Teaching and Examination Scheme of B.Sc.(IT) Programme

Teaching and Examination Scheme (B.Sc.(IT) Programme)

Effective from Year 2019-20

Semester-I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA401	Introduction to Computers	-	3	3	3	-	-	-	10	20	70	100
CA402	Introduction to Programming in C	4	3	7	7	10	20	70	15	15	70	200
CA403	Information Technology and Digital Electronics	4	-	4	4	10	20	70	-	-	-	100
CA404	Web Fundamentals	3	-	3	3	10	20	70	-	-	-	100
CA405	Foundation of Mathematics	3	-	3	3	10	20	70	-	-	-	100
HS101.01 C, HS102.01 C, HS105.01 C, HS109.01 C, HS110.01 C.	Liberal Arts	-	2	2	2	-		-	30		70	100
		14	08	22	22	400			300			700

Teaching and Examination Scheme (B.Sc.(IT) Programme)

Effective from Year 2019-20

Semester-II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA406	Operating System Concepts	4	3	7	7	10	20	70	15	15	70	200
CA407	Advanced Programming in C	-	3	3	3	-	-	-	15	15	70	100
CA408	Database Fundamentals	4	3	7	7	10	20	70	15	15	70	200
CA409	Computer Oriented Numerical Methods	3	-	3	3	10	20	70	-	-	-	100
CA410	Fundamentals of Commerce and Business Processes	3	-	3	3	10	20	70	-		-	100
HS126.01 C	Communication Skills - 1	2	-	2	2	30		70	-		-	100
		16	09	25	25	500			300			800

Teaching and Examination Scheme (B.Sc.(IT) Programme)

Effective from Year 2019-20

Semester-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA501	Programming the Internet	4	3	7	7	10	20	70	15	15	70	200
CA502	Fundamentals of Computer Networks	3	-	3	3	10	20	70	-	-	-	100
CA503	Object Oriented Programming Using C++	4	3	7	7	10	20	70	15	15	70	200
CA504	E-Commerce	3	-	3	3	10	20	70	-		-	100
CA505	System Analysis and Design	3	-	3	3	10	20	70	-		-	100
HS122 C	Values and Ethics	2	-	2	2	30		70	-		-	100
	University Elective-I **	-	2	2	2	-		-	30		70	100
		19	08	27	27	600			300			900

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA224 Introduction to Web Designing** course for others.

University Electives

No	Course Code	Course Name	Department/Faculty
1	CE281.01	Art of Programming	Engineering
2	CL281.01	Environmental Sustainability & Climate Change	Engineering
3	EE283	Python for Electrical Engineering	Engineering
4	IT281.01	ICT Resources & Multimedia	Engineering
5	ME281.01	Engineering Drawing	Engineering
6	PH233.01	Fundamentals of Packaging	Pharmacy
7	PD260.01	Basic Laboratory Techniques	Applied Science
8	NR251.01	First Aid & Life Support	Nursing
9	PT191.01	Health Promotion & Fitness	Physiotherapy
10	BM231	Banking & Insurance	Management
11	EC281.01	Introduction to MATLAB Programming	Engineering
12	PD261	Astrophysics, Space and Cosmos-I	Applied Science

Teaching and Examination Scheme (B.Sc.(IT) Programme)

Effective from Year 2019-20

Semester-IV

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA506	Data Structures & Algorithms	4	3	7	7	10	20	70	15	15	70	200
CA507	Advanced Database System	-	3	3	3	-	-	-	15	15	70	100
CA508-512	Elective – I	4	3	7	7	10	20	70	15	15	70	200
CA513	Management Information System	3	-	3	3	10	20	70	-		-	100
HS133 C	Creativity, Problem Solving and Innovation	-	2	2	2	-		-	30		70	100
CL142.01	Environmental Sciences	-	2	2	2	-		-	30		70	100
	University Elective-I I**	-	2	2	2	-		-	30		70	100
		11	15	26	26	300			600			900

****** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA225 Programming the Internet course** for others**.**

Elective-I

1. CA508 Graphics Designing
2. CA509 Open Source Technology
3. CA510 Multi Paradigm Scripting
4. CA511Foundation of Data Science
5. CA512 Fundamentals of Embedded Programming

University Elective – II

No	Course Code	Course Name	Department/Faculty
1	EC282.01	Prototyping Electronics with Arduino	Engineering
2	CE282.01	Web Designing	Engineering
3	CL282.01	Basics of Environmental Impact Assessment	Engineering
4	EE286	Computer Programming for Electrical Engineering	Engineering
5	IT282.01	Internet Technology & Web Design	Engineering
6	**6** ME282.01	Material Science	Engineering
7	PH238.01	Cosmetics in Daily Life	Pharmacy
8	NR261.01	Life Style Diseases & Management	Nursing
9	PT192.01	Occupational Health & Ergonomics	Physiotherapy
10	BM241	Health Care Management	Management
11	PD262	Astrophysics, Space and Cosmos-II	Applied Science

Teaching and Examination Scheme (B.Sc.(IT) Programme)

Effective from Year 2019-20

Semester-V

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA601	Object Oriented ProgrammingThrough JAVA	4	3	7	7	10	20	70	15	15	70	200
CA602	Application development using .NET Framework	4	3	7	7	10	20	70	15	15	70	200
CA603-607	Elective - II	4	3	7	7	10	20	70	15	15	70	200
CA608	Object Oriented System Analysis and Design	-	3	3	3	-	-	-	30		70	100
HS124.01 C	Professional Communication	-	2	2	2	-	-	-	30		70	100
		12	14	26	26	300			500			800

Elective-II

1. CA603 Fundamentals of Multi-Media
2. CA604 Open Source Framework - I
3. CA605 Scripting Framework - I
4. CA606 Advanced Data Science
5. CA607 Embedded Programming Using Arduino

Teaching and Examination Scheme (B.Sc.(IT) Programme)

Effective from Year 2019-20

Semester-VI

<table>
<tr><th rowspan="4">Course Code</th><th rowspan="4">Course Title</th><th colspan="4">Teaching Scheme</th><th colspan="7">Examination Scheme</th></tr>
<tr><th colspan="3">Contact Hours</th><th rowspan="3">Credit</th><th colspan="3">Theory</th><th colspan="3">Practical</th><th rowspan="3">Total</th></tr>
<tr><th rowspan="2">Theory</th><th rowspan="2">Pract</th><th rowspan="2">Total</th><th colspan="2">Internal</th><th rowspan="2">External</th><th colspan="2">Internal</th><th rowspan="2">External</th></tr>
<tr><th>Case Study</th><th>Tests</th><th>Term work</th><th>Tests</th></tr>
<tr><td>CA609</td><td>Minor Project</td><td>-</td><td>12</td><td>12</td><td>12</td><td>-</td><td>-</td><td>-</td><td>50</td><td>50</td><td>200</td><td>300</td></tr>
<tr><td>CA610</td><td>Mobile Application Development</td><td>4</td><td>3</td><td>7</td><td>7</td><td>10</td><td>20</td><td>70</td><td>15</td><td>15</td><td>70</td><td>200</td></tr>
<tr><td>CA611-615</td><td>Elective - III</td><td>3</td><td>3</td><td>6</td><td>6</td><td>10</td><td>20</td><td>70</td><td>15</td><td>15</td><td>70</td><td>200</td></tr>
<tr><td>HS134 C</td><td>Contributor Personality Development</td><td>-</td><td>2</td><td>2</td><td>2</td><td colspan="2">-</td><td>-</td><td colspan="2">30</td><td>70</td><td>100</td></tr>
<tr><td colspan="2"></td><td>07</td><td>20</td><td>27</td><td>27</td><td colspan="3">200</td><td colspan="3">600</td><td>800</td></tr>
</table>

Elective-III

1. CA611 Advanced Multi-Media
2. CA612Open Source Framework - II
3. CA613 Scripting Framework - II
4. CA614 Data Mining & Analytics
5. CA615 Embedded Programming Using Raspberry Pi

DETAIL SYLLABUS OF FIRST SEMESTER OF B.SC.(IT)

Teaching and Examination Scheme (BSc.(IT) Programme)

Effective from Year 2019-20

Semester-I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA401	Introduction to Computers	-	3	3	3	-	-	-	10	20	70	100
CA402	Introduction to Programming in C	4	3	7	7	10	20	70	15	15	70	200
CA403	Information Technology and Digital Electronics	4	-	4	4	10	20	70	-	-	-	100
CA404	Web Fundamentals	3	-	3	3	10	20	70	-	-	-	100
CA405	Foundation of Mathematics	3	-	3	3	10	20	70	-	-	-	100
HS101.01 C, HS102.01 C, HS105.01 C, HS109.01 C, HS110.01 C.	Liberal Arts	-	2	2	2	-		-	30		70	100
		14	08	22	22	400			300			700

CA401: Introduction to Computers
(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: Microsoft Word, Excel, Power Point and Access will be taught practically with its real time use. MS Word will be explained to produce and organize written material, correspondence and so on. Excel will be explained which is very popular business productivity application for the management and manipulation of data. With the right training and understanding of Excel, businesses and individual users can unlock the world of opportunities that this powerful business application offers. PowerPoint will be taught to create and show slides to create a professional presentation. Microsoft Access will be taught to create tables and queries.

Outline of the Course:

Week No.	Content
1	Introduction to Word Processing and MS word: Crating, saving and opening word documents, Menus, Shortcuts, Toolbars, Ruler Menus, Scroll bar, creating and editing a document, Using help, Setting Font Styles, Paragraph styles and Page styles, setting document styles
2	Insertion of Header & Footer, Word Art, Chart, Shapes, smart art
3	Creating tables, Drawing and formatting pictures and image, Spell check, Macros, Mail merge, Security, Printing documents, Formatting page and Setting Margins
4	Worksheet Basics: Creating worksheet, entering data into worksheet, heading information, data text, and alphanumeric values, saving & quitting worksheet.
	Opening and moving around in an existing worksheet. Toolbars and Menus, Keyboard Shortcuts.
5	Working with single and multiple workbooks: copying, renaming, moving, adding and deleting, copying entries and moving between workbooks, Working with formulae & cell referencing, performing auto sum, copying formula, absolute & relative addressing
6	Editing & Formatting: creating, editing and selecting ranges. Formatting of Worksheet: auto format, changing alignment, character styles, column width, date format, borders & colors and currency sign. entering and erasing data, resizing rows and columns, conditional formatting, adding comments.

7	Excel forms, graphs and charts: data entry forms using Excel, using wizards, various charts type, formatting grid lines & legends, previewing & printing charts
8	Basics font and paragraph formatting, creation of presentation, preparation of slides, editing data and images in slide, inserting image, chart, picture, shape and smart art & clip art
9	Providing aesthetics and animation, theme, slide manipulation and slide show setup
10	Header and footer, speaker notes, handouts, an outline, presentation of the Slides and templates, print preview and print
11	Feature of MS Access, understanding database concepts, creating database, create relationship, look-up fields, enumeration type, creating form and simple control
12	Display data, Queries: join two tables, outer join and group by, query with user criteria, Generate report

Total Hours (Practical): 36

Core Books:

1. Microsoft Word 2016 Step By Stepby Joan Lambert, Microsoft Press
2. Microsoft Excel 2016 Step By Stepby Joan Lambert, Microsoft Press
3. Microsoft Power Point 2016 Step By Stepby Joan Lambert, Microsoft 4. Microsoft Access 2016 Step By Stepby Joan Lambert, Microsoft

Reference Books:

1. Mastering Microsoft Excel Functions and Formulasby , Webtech Solutions Inc. 2. Microsoft Powerpoint 2013 Step by Step by Cox, Microsoft

Web References

1. http://www.gcflearnfree.org/word2016/[For MS Word]
2. https://www.techonthenet.com/excel/tutorial2016_complete.php [For Microsoft Excel] 3.http://www.gcflearnfree.org/powerpoint2016/ [Microsoft Power Point]
4. http://www.investintech.com/articles/mscustomization/[For Word, Excel, Power Point]
5. http://in.pcmag.com/office-suites/34995/feature/100-essential-tips-for-microsoftoffice2010[For Tips and Tricks of MS Office]
6. https://www.tutorialspoint.com/ms_access/index.htm[MS Access]

Course Outcomes: Upon successful completion of the course, the students will

C01 : To work with the basic features of MS Word, to compose, format and edit a word document

C02 : To work with the advanced formatting features of MS Word like table of content, table, illustration, coverpage and mail merge.

C03 : To work with the basic features of MS Excel, to compose, format and edit an excel workbook with advanced features like filtering, pivot table, chart, in-built functions and data analysis & organization.

C04 : To learn the creation of professional MS PowerPoint presentation by Providing aesthetics, animation& transition and multi-media.

C05 : To learn the basic of MS Access with query handling

Course Outcomes Mapping:

Week No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1-2	MS Word	✓	✓			
3-4	MS Word	✓	✓			
5-6	MS Excel			✓		
7-8	MS Excel			✓		
9-10	MS PowerPoint				✓	
11-12	MS Access					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	1	1	2	2	-	3	-	1
CO2	-	1	2	2	-	3	3	-
CO3	-	1	-	2	2	3	3	-
CO4	-	1	-	2	-	3	3	-
CO5	-	2	2	3	2	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA402: Introduction to Programming in C

(Marks 200)

Contact Hours: 07

Pre- requisite: None.

Methodology & Pedagogy: During theory lectures the emphasis will be given on theoretical concepts of flow charts and algorithms. Illustrations of certain real world problem, which are to be solved using computers, will be discussed. Structure of C programming language and different elements C programming will be explained in lectures. During Practical sessions, students will be required to Develop Computer programs in "C" in order to solve moderate size real world problems.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Introduction to Algorithms and Programing Languages	07	36
2	Introduction to C	08	
3	Decision control and looping statements	10	
4	Arrays	08	
5	Strings	07	
6	Functions	08	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total:84

Detailed Syllabus:

Unit – I: Introduction to algorithms and Programing languages Hours : 07

Introduction to algorithm, Key features of algorithm, Introduction to flow charts, Significance of flow chart, Advantages of flowchart, limitation of flow chart, Compiler, Interpreter, Programming languages

Unit – II: Introduction to C Hours: 08

Background of C, characteristics of C, Structure of a C Program, writing the first C program, Files used in C program, Compiling and Executing C program, Using Comments, Keywords, Identifiers, Basic data types in C, Variables, Constants, Input / Output statements in C, Operators in C

Unit – III: Decision control and looping statements Hours :10

Conditional Branching Statements, Iterative statements, Nested loops, Break, Continue and goto statements

Unit – IV: Arrays Hours : 08

Declaration of Arrays, accessing elements of the array, storing values in Arrays, Calculating the length of the array, Operation that can be performed on Arrays, two-dimensional Arrays

Unit – V: Strings Hours : 07

Introduction, Reading Strings, Writing Strings, String operations, String manipulation Functions, Arrays of Strings

Unit – VI: Functions Hours : 08

Introduction, Function declaration, Function definition, Function call, return statement, Passing parameters to the functions, storage classes, recursion

Core Books:

1. ReemaThareja:Introduction to C Programming,Oxford University press, 2012.
2. ReemaThareja: Computer Fundamentals and Programming in C,Oxford University press, 2012.

Reference Books:

1. Herbert Schildt: The complete reference C, Fourth edition, Tata McGraw Hill,2000
2. E. Balagurusamy: Programming in C, 6th Edition, 2012
3. YashwantKanetkar: Let us C, 13th Edition, BPB publication, 2013.
4. Brian W. Kernighan and Dennis M. Ritchie: The C Programming Language,2ndEdition,PHI.

Web References:

1. http://www.programiz.com/c-programming[For C programming]
2. http://cprogrampracticals.blogspot.in/p/special-c-programs.html[For C programming practical]
3. http://cprogramsblog.blogspot.in/ [For examples of C Programs]
4. http://www.tutorialspoint.com/cprogramming/[For C tutorials]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Understand about algorithm and flowchart with logical thinking with the ability to analyze a problem.
C02 : Learn about the structure of C program with demo and elements like variables, constant with the knowledge of operators used in C.
C03 : Gain the knowledge of different conditional and iterative statements.
C04 : Understand about working with Arrays with numbers and characters as well as manipulating with them.
C05 : Learn and exploring User defined functions.

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to algorithms and Programing languages	✓				
2	Introduction to C		✓			
3	Decision Control and Looping Statements			✓		
4	Arrays				✓	
5	Strings				✓	
6	Functions					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	1	2	2	-	2	3	-	3
CO2	3	3	-	3	3	1	2	-
CO3	-	2	-	3	-	3	-	2
CO4	-	2	-	1	-	-	2	-
CO5	2	-	2	3	1	1	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA403: Information Technology and Digital Electronics

(Marks 100)

Contact Hours: 04

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures foundations of information technology related concepts will be introduced to students. Students will be made familiar with applications, societal aspects and emerging trends related to information technology. Basic understanding of concepts and techniques used in digital electronics will be given to the students in form of theory sessions and case studies.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours
		Theory
1	Basics of Information Technology	05
2	Data Storage and organizations	08
3	Applications of IT and Emerging Trends and Societal Impact of IT	09
4	Computer Operation and Number System	08
5	Boolean Algebra and Gates Network	09
6	Digital Logic Circuits and Digital Components	09

Total Hours (Theory): 48
Total: 48

Detailed Syllabus:

Unit – I:Basics of Information Technology Hours: 05
Introduction, Types of Data, Simple Model of a Computer,CPU, I/O Devices,Data Processing Using Computer, Desktop Computer, Interrupt Controller

Unit – II: Data Storage and OrganizationsHours: 08
Storage Cell, Physical Devices Used as a Storage Cell, Random Access Memory, Read Only Memory,

Secondary Storage, Hard Disk,Compact Disc Read Only Memory, Archival Store and calculation

Unit – III: Applications of IT and Emerging Trends and Societal Impact of ITHours: 09

Business Applications, On-Line Applications, Real-Time Applications

Introduction to Emerging Trends in ICT, Ecommerce, Electronic Data Interchange, Mobile Communication, Bluetooth, Global Positioning System, Infrared Communication System, Smart Card, Imminent Technology

Introduction to Societal Impact, Social Use of World Wide Web, Privacy, Security and Integration of Information,Disaster Recovery, Intellectual Property Right, Career in IT

Unit – IV: Computer Operation and Number System Hours: 08

Electronic digital computers, Application of computers to problems, Basic components of a digital computer, construction of memory, instructions, programming systems, Decimal system, disable devices, counting in the binary system, binary addition and subtraction, binary multiplication and division, converting decimal numbers to binary, negative numbers, use of complements to represent negative numbers, binary number complements, binary coded decimal number representation, octal and hexadecimal number systems.

Unit – V: Boolean algebra and Gates Network Hours: 09

Fundamental concepts of Boolean algebra, logical multiplication, AND gates and OR gates, complementation and inverters, evaluation of logical expressions, basic laws of Boolean algebra, De Morgan's theorem, basic duality of Boolean algebra, derivation of Boolean expression, interconnecting gates, sum of products and product of sums, NAND and NOR gates, don't care conditions, design using NAND and NOR gates.

Unit – VI: Digital Logic Circuits and Digital Components Hours: 09

Digital components, combinational circuits, flip flops, sequential circuits, integrated circuits, decoders, multiplexers, registers, shift registers, binary counters, memory unit.

Core Books:

1. V Rajaraman: Introduction to Information Technology, 2nd Edition, PHI Learning Private Limited, 2013.
2. Thomas C. Bartee: Digital Computer Fundamentals, Sixth Edition, Tata McGraw Hill Publishing, 2012.

Reference Books:

1. Turban,Rainer,Potter: Introduction to Information Technology, 2nd Edition,Wiley India, 2005.
2. PelinAksoy,LauraDenardis : Information Technology in Theory, Thomson Course Technology,2008.

3. Andrew S. Tanenbaum: Structured Computer Organization, Fourth Edition, Pearson Education, 2005.
4. Abert Paul Malvino and Jerald A. Brown: Digital Computer Electronics, Third Edition, Tata McGraw Hill Publishing, 2008.

Web References:

1. https://www.safaribooksonline.com/library/view/designingembeddedhardware/0596007558/ch01.html[Computer Architecture]
2. http://whatis.techtarget.com/definition/logic-gate-AND-OR-XOR-NOT-NAND-NOR-andXNOR[Logical Gates Tutorials]
3. https://www.tutorialspoint.com/computer_logical_organization/codes_conversion.htm[Code Conversion]
4. http://m.noteboy.in/NoteBoy_files/filless/8.%20Multiplexer%20and%20demultiplexer.pdf[MultiplexerDeMultiplexer]
5. http://www.edutechlearners.com/computer-fundamentals-p-k-sinha-free-pdf[E-Book]
6. https://sites.google.com/site/viveklpm/information-technology-inveterinaryscience/applications-of-information-technology[Information Technology Notes]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Understand the data, information and tool to process them.

C02 : Understand the concept of Information Technology and real applications of it.

C03 : Understand the Emerging trends and Societal Impact of Information Technology.

C04: Learn to convert one number to another number system.

C05: Learn the concepts of digital electronics required for the real application.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Basics of Information Technology	✓	✓			
2	Data Storage and organizations		✓			
3	Applications of IT and Emerging Trends and Societal Impact of IT			✓		
4	Computer Operation and Number System				✓	
5	Boolean Algebra and Gates Network					✓
6	Digital Logic Circuits and Digital Components					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	-	-	-	1	-	-
CO2	-	3	2	3	-	3	2	-
CO3	2	-	-	2	2	-	3	-
CO4	2	2	-	-	-	-	2	-
CO5	1	2	-	3	2	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA404: Web Fundamentals

(Marks 100)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: In order to achieve the course objectives**,** student emphasis will be given on the Introduction to the Internet including World Wide Web (WWW), services provided by the Internet and protocol used. They will learn to browse and search the websites using web browsers.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours
		Theory
1	Basics of the Internet	06
2	Internet Domain Name System	06
3	Internet Connectivity	07
4	Internet Services and Protocols	06
5	World Wide Web (www)	06
6	Internet Collaboration	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Basics of the Internet Hours: 06

Overview of Internet, Evolution of Internet, Overview of Intranet and Extranet, Internet reference models.

Unit – II : Internet Domain Name System Hours: 06

Overview of Domain Name system(DNS), IP Address, Uniform Resource Locator (URL), Absolute and Relative URL, Domain Name System Architecture, Domain Name Space, Name Server, working of DNS.

Unit – III: Internet Connectivity Hours: 07

Overview of Internet Service Providers (ISP), Working of Internet Service Providers(ISP), Different types of connection like Dial-up Connection, ISDN, DSL, Cable TV Internet connections, Satellite Internet connections, Wireless Internet Connections.

Unit – IV: Internet Services and Protocols Hours: 06

Overview of Services provided by the Internet, Communication Services, Information Retrieval Services, Web Services, Overview of Internet Protocols like Transmission Control Protocol (TCP), Internet Protocol (IP), User Datagram Protocol (UDP), File Transfer Protocol (FTP), Trivial File Transfer Protocol (TFTP).

Unit – V: World Wide Web(www) Hours: 06

Concept of www, Evolution of www, www architecture, concept of web page, linking web pages, web browsers, web servers, proxy servers, search engines and its components like Web Crawler, Database and Search Interfaces, overview of World Wide Web Consortium(W3C).

Unit – VI: Internet Collaboration Hours: 05

Email, Chatting, Instant Messaging, Video Conferencing, Video Sharing, Online Training, Online certification, Online Seminar, Webinar, Conferencing, social networking services provided over web and mobile, Remote Sharing Softwares.

Core Books:

1. John R. Levine,Margaret Levine Young:The Internet for Dummies(Paperback), 13th Edition, John Wiley & Sons, 2012.
2. Adesh K. Pandey: Internet Fundamentals (Paperback) ,SkKataria& Sons, 2010.

Reference Books:

1. Margaret Levine Young: Internet the Complete Reference, Osborne/McGraw-Hill, 2nd Revised edition, June 2002.
2. Rajkamal: Internet and web Technologies, Tata McGraw Hill, 2002.
3. C.S.Rayudu: E-Commerce-Business , Himalaya Publishing House Mumbai , 2008.
4. Raymond Greenlaw:Inline/Online: Fundamentals of The Internet & The World Wide Web, 2nd edition, McGraw-Hill Higher Education,2001.

Web References:

1. http://luc-it.blogspot.in/2013/05/internet-fundamental-applications.html[Internet Basics]
2. https://www.ntchosting.com/encyclopedia/internet/[Internet terms]
3. http://www.roseburg.k12.or.us/depts/tech/StaffDev/Int4Ed/Resources/www7.html[Basic Internet Tools]
4. http://www.gcflearnfree.org/internet101[Introduction to the Internet]

Course Outcomes: Upon successful completion of the course, the students will

C01 : Understand the need of the internet and its origin along with various internet related terms and references.

C02 : Learn about domain name systems as one of the core service of today and how they work.

C03 : Learn about different methods to connect with internet with their advantages and disadvantages.

C04 : Understand various services provided on the internet, including the web and learning about TCP/IP and UDP protocols that make internet work.

C05 : Get knowledge about origins of world wide web, its components and related concepts.

C06 : Insights on different methods for collaboration with help of internet and their applications.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Basics of Internet	✓					
2	Internet domain systems		✓				
3	Internet connectivity			✓			
4	Internet services and protocols				✓		
5	World wide web					✓	
6	Internet collaborations						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	2	-	2	-	1	2	-
CO2	-	-	1	-	-	1	-	2
CO3	2	-	-	2	-	-	-	-
CO4	-	-	1	-	1	-	-	1
CO5	-	-	1	-	-	2	-	-
CO6	-	-	-	2	-	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA405 Foundation of Mathematics

(Marks 100)

Contact Hours: 03

Pre-requisite: No additional Pre-requisite required

Methodology & Pedagogy: The emphasis during Lecture sessions will be on Understanding of Concepts rather than on complexities of Computational Techniques. Stress is also given on Visual perception through Mathematical Software. Relevant Illustrations will be provided from the Real World processes. Sufficient home assignments will be given to the students which will test their fundamentals and ability to relate Mathematical concepts with reality.

Outline of the course:

Unit No.	Title of the Unit	Minimum Numbers of Hours
		Theory
1	Set Theory	05
2	Functions	05
3	Matrix(Matrices)	06
4	Limit & Continuity	06
5	Differentiation & its applications	07
6	Integration & its applications	07

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit - I: Set Theory Hours:05

Introduction to set theory, Methods of representation of a set, Operations on set, Properties of set with logical proofs, Venn Diagram, Cartesian product of sets

Unit- II: Relations &FunctionsHours:05

Definition of relations, Types of relations, Function: Definition of Function, Classification of function, Domain and Range of function, Types of functions

Unit- III: Matrices and DeterminantsHours:06

Matrices, Matrix Operations, Properties of Matrix Operations, Determinants (up to order 3) and their properties, Cofactor expansion, The Inverse of a Matrix, The Rank of a Matrix and Applications.

Unit- IV: Limit& Continuity Hours:06

Limits: Definition of Limit, Evaluation Techniques

Continuity: Definition of Continuity, Conditions for the function to be Continuous, discontinuity

Unit- V: Differentiation & its applications Hours:07

Definition of differentiation, Rules of differentiation, Evaluation Techniques, Derivatives of Algebraic, trigonometric, Logarithmic, Explicit/Implicit function, Second order Derivative with example

Unit- VI: Integration & its application Hours:07

Definition of Integrations, Primitives of Standard Functions, Methods ofIntegration, Integration by parts, Applications of Integration: Area Bounded by the Curve

Core Books:

1. B.S.ShahPrakashan: Business Mathematics
2. S.Chand& Sons Publications
3. Thomas G. B. and R. L. Finney: Calculus and Analytical Geometry, 9th Ed., Addison Wesley,1996.
4. Erwin Kreyszig: Advanced Engineering Mathematics, 8th Ed., John Wiley & Sons, India, 1999

Reference Books:

1. Larsen & Marx: An Introduction to Mathematical Statistics and Its Applications, Third Ed., Prentice Hall, NJ, USA, 2001.
2. Wylie & Barrett: Advanced Engineering Mathematics, Mcgraw Hill pub.
3. Greenberg M D: Advanced Engineering Mathematics, 2nd ed., Pearson Education

Web References:

1. http://www.mathtutor.ac.uk/differentiation [For Differentiation & Integration]

2. http://www.maths.manchester.ac.uk/~mdc/old/1K1/DiscreteMaths.html [For Set Theory, Functions and Matrices]

Course Outcomes: Upon successful completion of the course, the students will be able to:

C01 : Formulate problems in the language of sets and perform various set operations. Draw and use Venn diagrams to solve practical problems.

C02 : Understand basics of relations and functions, types of relations and functions and classification of functions.

C03 : Understand basics of Matrix, performing various operations on matrix, types of matrices, applications of matrix (to find determinant and linear equation solutions)

C04 : Compute basic limits of functions. Understand the importance of limits to the process of differentiation and be able to compute the derivative of a simplefunctions. Compare and contrast the ideas of continuity and differentiability.

C05 : Use the basic integration rules to find indefinite integrals, Use the Fundamental Theorem of Calculus to evaluate definite integral and its application as to find the area under the curve.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Set Theory	✓				
2	Functions	✓	✓			
3	Matrix(Matrices)			✓		
4	Limit & Continuity				✓	✓
5	Differentiation & its applications				✓	✓
6	Integration & its applications					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	1	2	3	1	1	1	1	1
CO2	3	2	3	1	1	1	2	1
CO3	3	2	3	1	1	1	2	1
CO4	2	2	3	1	1	2	2	1
CO5	2	1	3	2	1	2	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS101.01 C: PAINTING

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS101.01 A/B/C/D/E/F/G/H	Painting	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction to Painting** • *An Introduction to Painting* • *Principles of Composition* • *Medium and Techniques of Painting* • *History of Painting: Folk Indian Painting / Western Painting* • *2D and 3D Painting*	2
2	**Drawing from Nature and Object** • *Objects of Drawing: Nature and Manmade / Artificial Objects* • *Drawing Still / Live Objects* • *Drawing from Memory* • *Drawing from Life*	4
3	**Colour Design and Colour Value** • ***Color Theory****:* *Color wheel (primary/secondary, complementary), transparency/opacity, hue, value (intensity, brightness), chroma (saturation, purity) & temperature (warm/cold)* • ***Color Contrast & Attributes****:* *Interaction, harmony, psychology/mood, culture & expression* • ***Media Characteristics & Surfaces****:* *Acrylic, oil, paper, wood & canvas (primed/unprimed)*	6

4	**Composition and Perspective** • ***Composition:*** *Space, movement, balance, asymmetry, rhythm, shapes, proportion & lighting* • *Perspective: An approximate reproduction* • ***Types of Perspectives:*** *Linear Perspective, One-point Perspective, Two-point Perspective, Three-point Perspective, Four-Point Perspective*	6
5	**Figure Drawing and Proportion** • *Proportions of the Human Body* • *Three views –* ***Anterior (front), Lateral (side) and Posterior (back)*** • *Fundamental Proportion – The Big Three*	4
6	**Sketching** • *Sketching and Freehand* • *Sketching Techniques* • *Sketch and Drawing Medium*	4
7	**Contemporary Issues in Painting** • *Contemporary Indian Art* • *Pioneers of Contemporary Indian Art* • *Contemporary Issues in Painting*	4
	Total Hours	**30**

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15

4	Attendance	-	05	05
			Total	**30**

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	**Component**	**Number**	**Marks per incidence**	**Total Marks**
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course, the student would :

CO1 have cultivated a sense of creativity.

CO2 be appreciative of art history, art criticism and aesthetics.

CO3 be able to recognize the elements of arts in painting.

CO4 have better cognizance and association of meaning of colors, shapes, and composition.

CO5 be able to acknowledge the principles of painting as in design and colors, concept, medium and formats.

CO6 have instantaneous painting experience about designing, lights, shades and colors and such other important aspects.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	3	3	-
CO5	-	-	-	-	-	3	-	-
CO6	-	-	-	-	-	3	-	-

Correlation levels 1, 2 or 3 are as defined below:1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS102.01 C: PHOTOGRAPHY

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS102.01 A/B/C/D/E/F/G/H	Photography	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction to Photography** • *Art, Design and Visualization* • *Basics of Photography and Various Types of Photography* • *Basics of Post Production* • *A Brief History of Photography: Early Experiments and Later Developments*	03
2	**Camera and Operating System** • *Role of Camera in the Photography* • *Types of Camera* *Pin-hole, box, folding, large and medium format cameras, single lens reflex (SLR) and twin lens reflex (TLR), miniature, subminiature and instant camera* • *Principal Parts of Photographic Camera* *Lens, Aperture, Shutters, various types and their functions, focal plane shutter and in-between the lens shutter, shutter synchronization, self-timer* • *Types of Lenses* *Single (meniscus), achromatic, symmetrical and unsymmetrical lenses, telephoto, zoom, macro, supplementary and fish-eye lenses* • *Different Models of Camera, their Features and Operating Systems* • *Camera and Size of the Image, Speed and Power of Lens*	05

3	**Light and Shade** • *Reflection and refraction of light* • *Dispersion of light through a glass prism, lenses* • *Colour Filters:* *Different kinds, Red, yellow, green, neutral density, half filters, filter factor, colour correction filter* • *Photographic Light Sources:* *Natural source, the Sun, nature and intensity of the sunlight at different times of the day, different weather conditions* • *Artificial light sources:* *Nature, intensity of different types of light sources used in photography namely; (i) Photo flood lamp, (ii) Spot light, (iii) Halogen lamp, Barn doors and snoot, lighting stands* • *Flash unit: Bulb flash and Electronic flash, main components, electronic flash units, studio flash, slave unit, multiple flash, computer flash, x-contact, exposure table*	**10**
4	**Composition** • *Different kinds of image formations* • *Principal focus and focal length of the lens* • *Depth of field, angle of view and perspective* • *Perspective and composition* • *Rules of composition*	**09**
5	**Contemporary Issues in Photography** • *Present Day Photography* • *Contemporary Photographers and their Contributions* • *Major Issues in Contemporary Photography*	**03**
	Total Hours	**30**

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
			Total	**30**

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course, the student would:

CO1 Understand, appreciate and demonstrate innovative approach, beauty and acute acumen in the area of photography

CO2 Develop photography skills and become familiar with the functions and importance of the visual elements of nature and artificial objects

CO3 Become independent thinkers who will contribute inventively and critically to culture through the making of art photography

CO4 Have thorough understanding and acute sense of light and shade, composition, and presentation of a piece of an art

CO5 Experiment and Represent the cultivated sense and skills in Photography to the mass

CO6 Prepare an impressive portfolio encompassing holistic approach to art and other the areas of study

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	3	-	-
CO5	-	-	-	-	-	3	2	-
CO6	-	-	-	2	-	3	2	-

Correlation levels 1, 2 or 3 are as defined below:1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS105.01 C: MEDIA AND GRAPHIC DESIGN

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS105.01 A/B/C/D/E/F/G/H	Media and Graphic Design	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction to Media and Graphic Design** • *Creating Art, Art in Context and Art as Inquiry* • *History of Graphic Design* • *Constructional, Representational, and Simplification Drawing*	03
2	**Layout and Design** • *Layout, Design and Aesthetics* • *Elements of Design* • *Principles of Design: Harmony, Balance, Rhythm, Perspective, Emphasis, Orientation, Repetition and Proportion* • *Impact/function of Design* • *Indigenous design practices* • *Role of design in the changing social scenario*	07
3	**Form and Space** • *Types of Forms: Man-made, Nature* • *Types of Space: Negative and Positive* • *Composition of Form and Space to create Layout* • *Exploring Creativity*	06
4	**Computer Graphics** • *An Introduction to Graphic Software* • *Flash, Coreldraw, Illustrator and Photoshop*	04

	• *Pre-press Process*	
5	**Fonts** • *Construction of Type* • *Anatomy of Type* • *Visual Language* • *Creating Logo and Symbol*	**04**
6	**Basic Print Media** • *An Introduction to Press and its Development Phases* • *Types of Press* • *Types of Printing Technologies* • *Post-press Processes*	**03**
7	**Contemporary Issues in Graphic Design** • *Present Day Graphic Designs* • *Contemporary Designers and their Contribution* • *Major Contemporary Issues in Graphic Design*	**03**
	Total Hours	**30**

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
			Total	**30**

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course, the student would:

CO1 have cultivated a sense of creativity.

CO2 be appreciative of art and designs, art criticism and aesthetics.

CO3 be able to recognize the elements of arts in graphic design.

CO4 have better cognizance and association with the meaning of designs, shapes, colors, print and medium.

CO5 be able to design graphics using computer softwares like Photoshop, CorelDraw, and Illustrator.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	3	-	-
CO5	-	-	-	-	2	3	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS109.01 C || DRAMATICS

I. Credits and Schemes

Sem.	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours / Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS109.01 A/B/C/D/ E/F/G/H	Dramatics	02	02	--	--	30	70	100

II. Course Outline

Module	TITLE / TOPIC	Classroom Contact Sessions
1	**Introduction to Drama** • Introduction to performing arts • Drama - An art, a socializing activity, & a way of learning • Form of Drama • Elements of Drama • Types of Drama	**06**
2	**History of Drama and Contemporary Theatre** • Important world dramatists & drama–from Greek to modern • Evolution of contemporary theatre in the context of developments in Indian theatre • Major Movements in Drama	**06**
3	**Theatre Design and Techniques** • Theatre Architecture • Stage craft: Set, light, costume, make up, sound, props • Theatre techniques: from selection of script to final performance	**06**
4	**Technicalities of Stage Performance** • Selection of plot and character • Improvisation • Movement • Voice, Speech, Imagination • Character Development • Scene Enactment	**08**
5	**Contemporary Trends in Drama**	**04**

	• New Tendencies in theatre • Drama and Society • Using drama for Social Change and Education	

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

V. Internal Evaluation:

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				**30**

VI. External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				**70**

Course Outcome (COs):

After completion of the course, the student would:

CO1 be aware about the concept of performing art and its nuances.

CO2 display a working knowledge of historic of drama, its development and current trends in dramatics.

CO3 demonstrate skills in the technical/design preparation and execution of a theatre performance.

CO4 demonstrate the ability to work collaboratively.

CO5 develop essential transferable skills in various relevant areas of the theatre.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	3	2	-
CO3	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	3	-	-
CO5	-	-	-	-	-	3	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS110.01 C || CONTEMPORARY DANCE

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
1	HS110.01 A/B/C/D/E/F /G/H	Contemporary Dance	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title / Topic	Classroom Contact Sessions
1.	**Introduction to dance** • Dance as a Performing Art • Dance as a Medium of Expression • History and Development of Dance	4
2.	**Types of Dance** • Western dance and classical dance • Salsa, rumba, hip hop, tap dance, belly dance, etc. • Indian Classical Dance forms: Odissi, Bharatanatyama, Kathak, Kathakali, Kuchipudi etc. • Other Regional dance forms in India	6
3.	**Basic Elements of Dance** • Movements of different parts of a body for Expression • Concepts of: Nritya, Laya and Taal	4
4.	**Technical Skills in Professional Contemporary Dance** • Dance technique: alignment, balance, co-ordination, flexibility and control • Expressive / presentation skills: Dynamic energy, physical engagement with the given material and stage, etc. • Skills and processes of rehearsal and production: physical energy, stamina and athleticism	6

	• Musicality: clarity of timing and phrasing	
5.	**Contemporary Trends in Dance :** • Prevalent trends and techniques in contemporary dance • Future trends in contemporary dance form • On Stage Performance	**10**
Total Hours		**30**

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
			Total	**30**

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sr. No	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course, the student would:

CO1 be able to develop ability to express through the form of dance.

CO2 have enhanced aesthetic sensitivity.

CO3 have improved concentration, mental alertness, quick reflex action, and physical agility.
CO4 be able to express a natural way human feelings and expressions by creating harmony.
CO5 be able to deliver contemporary dance performance.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	3	2	-
CO5	-	-	-	-	-	3	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

DETAIL SYLLABUS OF SECOND SEMESTER OF B.SC.(IT)

Teaching and Examination Scheme (BSc.(IT) Programme)

Effective from Year 2019-20

Semester-II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA406	Operating System Concepts	4	3	7	7	10	20	70	15	15	70	200
CA407	Advanced Programming in C	-	3	3	3	-	-	-	15	15	70	100
CA408	Database Fundamentals	4	3	7	7	10	20	70	15	15	70	200
CA409	Computer Oriented Numerical Methods	3	-	3	3	10	20	70	-	-	-	100
CA410	Fundamentals of Commerce and Business Processes	3	-	3	3	10	20	70	-		-	100
HS126.01 C	Communication Skills - 1	2	-	2	2	30		70	-		-	100
		16	09	25	25	500			300			800

CA406 : Operating Systems Concepts

(200 Marks)

Contact Hours: 07

Pre-requisite: None.

Methodology & Pedagogy: In order to achieve the course objectives, students will first be introduced to the basic operating system concepts and basic functions. During practical sessions, the students will be required to use DOS and UNIX commands to understand the system properties and to write script files.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1.	Basic Concepts and Usage of OS	06	36
2.	Process Management	08	
3.	CPU Scheduling	08	
4.	Memory Management	06	
5.	Operating System Commands ,Editors and Filters	10	
6.	Shell Programming	10	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit - I: **Basic Concepts and Usage of OS Hours: 06**
Operating System Overview: OS objectives and functions (OS as User/Computer interface, OS as Resource Manager), Evolution of OS, Operating System services, User Operating System Interface, System Call, Operating System Structure, Operating System Design and issues involved – Interrupts, Interrupt Processing, Interrupt Vectors

Unit - II: **Process Management Hours: 08**

Process Control Block, Process States-A two state process model, A five state process model, A seven state process model, Process Operations- creation, termination and suspension, Process Vs. Thread, Inters
process communication- independent and cooperative process.

Unit - III: **CPU Scheduling Hours: 08**

Types of scheduling - long, medium, short, I/O Scheduling Criteria, Scheduling Algorithms - FIFO, Round Robin, Priority Scheduling, SPN, SRN, Feedback

Unit – IV : **Memory Management Hours: 06**

Memory Management Requirements, Partitioning, Paging, Virtual memory

Unit - V: Operating System Commands, Editors and Filters Hours: 10

Basic DOS commands, Batch file**,** History and Features of Unix OS, Directory structure, Basic commands – Metacharacter, Shell Variable, Command Substitution, Recording Script, Navigating File System, Unix Kernel and Shell, Editors: vi, emacs, pico, Simple Filters and pipes, standard Input, Standard output and Standard Error, Advance Filter – Grep

Unit - VI: Shell Programming Hours:10

Shell Script, Shell environment, Making Script interactive, Command Line Argument, Evaluating Expression, Operator, Control Statement, Looping statement

Core Books:

1. William Stallings: Operating Systems Internals and Design Principles, 5th Edition, PHI, 2005.
2. Sumitabha Das: Unix concepts & application, 4th Edition, Tata McGraw Hill, 2010.
3. Kenneth Rosen, Douglas Host, James Farber and Richard Rosinski: The Complete Reference, Tata McGraw Hill, 1999.

Reference Books:

1. Silberschatz: Operating System Concepts, 5th Edition, John Wiley & Sons (ASIA) Ltd., 2008.
2. Mark G. Sobell: A Practical Guide to Linux, Pearson Education, 1997.
3. K.J.George, Operating System Concepts and Principles, Sroff Publishers, 2003

Web References:

1. http://www.nondot.org/sabre/os/articles [A useful collection of documents and papers on a wide range of OS topics]
2. http://www.ugu.com/sui/ugu/warp.ugu [Excellent source of UNIX information]
3. http://www.tutorialspoint.com/unix/index.htm [Unix Tutorials]
4. http://williamstallings.com/OS/Animation/Animations.html [Animations of Operating Systems Concepts]

5. http://placement.freshersworld.com/power-preparation/technical-interviewpreparation/osinterview-questions-23351 [Operating System questions and answers]
6. http://www.ics.uci.edu/~ics143/lectures.html [Lecture notes of OS]

Course Outcomes: Upon successful completion of the course, the students will

C01 : Understand operating system overview, Functions and Services

C02 : Learn about various states of process models with reasons for process creation and termination.

C03 : Recognizing different CPU Scheduling algorithms with types of scheduling

C04 : Learn how memory is managed internally with Partitioning, Paging and Segmentation

C05 : Hands on with Shell Scripts by implementing programming elements.

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		C01	**C02**	**C03**	**C04**	**C05**
1	Basic Concept and Usage of OS	✓				
2	Process Management		✓			
3	CPU Scheduling			✓		
4	Memory Management				✓	
5	Operating System Commands, Editors and Filters					✓
6	Shell Programming					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	2	-	3	3	-	3
CO2	3	3	-	3	3	3	2	2
CO3	-	2	3	3	-	3	3	-
CO4	-	2	-	-	2	-	3	3
CO5	3	-	2	3	-	3	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA407: Advanced Programming in C

(100 Marks)

Contact Hours: 03

Pre-requisite: Fundamental Concepts of Programming Language

Methodology & Pedagogy: During practical sessions the concepts structure, pointers and file management will be implemented in depth by considering real life systems by using file management.

Outline of the Course:

Week No.	Practical	Description
1.	Introduction to Pointers, Declaring Pointer Variables, Pointer Expressions and Pointer Arithmetic, Null Pointers, Generic Pointers, passing arguments to Function using Pointers	Students will understand computer's memory organization, pointer concepts and its implementation in programming.
2.	Pointers and Arrays, Passing an Array to a Function, Difference between Array name and Pointer, Pointers and Strings, Array of Pointers, Pointers and 2D Arrays	Students will be able to understand significance of Pointers using arrays.
3.	Introduction to Structures, Nested	Students will learn how to create their
	Structures, Arrays of Structures	own user-defined data types in different ways.
4.	Structures and Functions, Self-Referential Structures	Students will learn implementation of Structures with functions.
5.	Union, Arrays of Union Variables, Unions Inside Structures, Enumerated Data Types	Students will learn how to create their own user-defined data types in different ways.

6.	Introduction to Files, Using Files in C, Read Data from Files, Writing Data to Files, Detecting the End-of-File, Error Handling During File Operations	Students will able to perform input/output operations on file.
7.	Accepting Command Line Arguments, Functions for Selecting a Record Randomly, remove(), Renaming the File, Creating a Temporary File	Students will learn managing file in C.
8.	Introduction to Preprocessor directives, Types of Preprocessor Directives, #define, #include, #undef, #line, Pragma Directives, Conditional Directives, #error Directive, Predefined Macro Names	Students will learn the usage of preprocessor directives and macros in C programming.
9.	Introduction, Linked Lists Versus Arrays, Memory Allocation and Deallocation for a Linked List	Students will learn concept of Dynamic Memory Allocation using Linked lists.
10.	Different Types of Linked Lists: Singly Linked List, Circular Linked List, Doubly Linked List	Students will learn different types of linked lists.
11.	Stacks, Array Representation of Stacks, Operations on a Stack, Infix, Postfix, and Prefix Notations	Students will understand and implement derived data types.
12.	Evaluation of an Infix Expression, Convert Infix Expression to Prefix Expression, Applications of Stack, Queues, Operations on a Queue	Students will learn applications of Stack and operations of Queue.

Core Books:

1. ReemaThareja. “Introduction to C Programming”, Oxford University Press.

Reference Books:

1. ReemaTharej. “Computer fundamentals and Programming in C”, Oxford University Press.
2. E. Balagurusamy. “Programming in ANSI C”, Sixth Edition, McGraw Hill Education.

3. PradipDey and ManasGhosh. “Programming in C”, Second Edition, Oxford University Press.
4.Brian W. Kernighan and Dennis M. Ritchie: The C Programming Language,2nd Edition,PHI

Web References:

1. https://www.codingunit.com/c-tutorial-how-to-use-pointers[For Pointers]
2. http://www.tutorialspoint.com/cprogramming/c_pointers.htm[For Pointers]
3. http://www.tutorialspoint.com/cprogramming/c_structures.htm[For Structures]
4. http://www.tutorialspoint.com/data_structures_algorithms/dsa_queue.htm[For stack & queue]
5. http://www.studytonight.com/data-structures/queue-data-structure[For data structures] 6. http://fresh2refresh.com/c-programming/[For entire C Programming Material]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : To understand basic of Pointer and operation on Pointers. Understand the concept of Dynamic memory management. Learn Programing-using Array with Pointer and Structure.

C02 : Learn operation on File. Read, write and Update file in C.

C03 : Understand basic data structures such as arrays, linked lists, stacks and queues.

C04 : To gain understanding of various Application implemented in C Language.

C05 : To learn various preprocessor directives.

Course Outcomes Mapping:

Week No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1-2	Pointer in C	✓		✓		
3-5	Structure, Union, and Enumerated Data Types	✓	✓	✓	✓	
6-7	Files	✓	✓			
7-8	Preprocessor Directives					✓
9-10	Linked Lists			✓	✓	
11-12	Stacks and Queues			✓	✓	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	1	3	-	3	-	-	-
CO2	2	2	3	-	3	-	-	-
CO3	2	1	3	-	2	-	-	-
CO4	3	3	1	-	1	-	-	-
CO5	1	-	2	-	2	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA408: Database Fundamentals
(200 Marks)

Contact Hours: 07

Pre-requisite: Basic knowledge of working with computer.

Methodology & Pedagogy: During the theory lectures, concepts for data storage will be discussed. Also DBMS and RDBMS concepts will be discussed in depth. The database applications used in the real world will be discussed with necessary examples. During the laboratory hours students will implement the concepts that are discussed during lecture.

Outline of the Course:

Unit	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Database System	04	36
2	Data Modeling using Entity Relationship Model	08	
3	Relational Data Modeling	08	
4	Database Design Methodology	08	
5	Schema Definition, Constraints, Queries	10	
6	Advanced Query Processing	10	

Total Hours(Theory):48
Total Hours(Lab):36
Total:84

Detailed Syllabus:

Unit -I: Introduction to Database System Hours:04

Basic Concepts: data, database, database systems, database management system
Purpose and advantages of Database management system (over file systems)
Various data models–ER Model and Relational Model, Three level architecture, Structure of DBMS ,Database actors and workers

Unit -II: Data Modeling using Entity Relationship Model Hours:08

Entity Types, Entity Sets, Attributes, Keys, Relationship Types, Relationship Sets, Roles and Structural Constraints, Weak Entity Types, ER Diagram–Notations and fundamentals, Conversion from ER Model to Relational Model

Unit -III: Relational Data Modeling Hours: 08

RelationalModelConcepts,RelationalModelConstraints,RelationalModelSchemas,Update Operation, Transactions, Dealing with Constraint Violations, Relational Algebra and Calculus, Database Migration

Unit -IV: Database Design Methodology Hours:08

Functional Dependency and Normalization for Database–Informal Design Guidelines for Relational Schemas, Functional Dependencies, Normal Forms (1NF, 2NF, 3NF,BCNF,4NF, 5NF), Codd Rules

Unit -V: Schema Definition, Constraints, Queries Hours: 10

Table Fundamentals **:**Basic Data types, Create Table Command, Viewing Data in the tables, sorting,Insert,DeleteandUpdateStatementsinSQL,Modifyingthestructureoftables,renaming table, truncating table, destroying table.
Data Constraints and Functions:- Pseudo columns, Nullvalues, TABtable, DUAL table Operators, Data constraints, Type of data constraints, Modifying constraints, working with data dictionary and use of USER_CONSTRAINTS
Functions–introduction, merits and demerits, types of functions
Numeric functions, Character functions, Date functions, Conversion functions, Aggregate functions

Unit -VI: Advanced Query Processing Hours: 10

Union, Intersect and Except Nested Queries – Introduction, Co-related Nested Queries, Set Comparison Operators
Aggregate Operators, Group By clause, having clause, Order by clause
Null Values– Comparison using Null Values, Logical Connectivity–AND ,OR and NOT ,Impact on SQLconstructs,Joins(InnerJoin,OuterJoin,SelfJoin,EquiJoin,CrossJoin)ComplexIntegrity Constraints in SQL– constraints over single table, domain constraints and distinct types
Creation and manipulation of database objects–indexes, views, sequences and synonym

Core Books:

1.RamezElmasri, ShamkantB.Navathe: Fundamentals of DatabaseSystems,5thEdition, Pearson Publication.
2. IvanBayross:SQL,PL/SQL–TheprogrammingLanguageOracle.
3. Ramkrishnan, Gehrke:DatabaseManagement Systems,3rdEdition,McGrawHillPublication.

Reference Books:

1. Silberschatz, Korth, Sudarshan :DatabaseSystem Concepts,5th Edition,McGraw Hill.
2. C.J.Date,aKannan,SSwaminathan:AnIntroductiontoDatabaseSystems,8th Edition, Pearson Education,(Equivalent Reading).
3. ScootUrban :Oracle9i, PL/SQL Programming, Oracle Press.
4. S. K. Singh:Database Systems: Concepts, Design and Applications, Pearson Education
5. Anjali, Amisha, Roopal and Nirav: Practice book on SQL and PL/SQL.
6. Leon and Leon: Database management Systems, VikasPublication.
7. PeterRob, Carlos Coronel: Database Systems: Design, Implementation and Management, 7th Edition, CengageLearning,2007.

Web References:

1. http://www.microsoftvirtualacademy.com/trainingcourses/databasefundamentals#?fbid=tbZ92pOp_Tt [Foroverallcourse]
2. http://www.ntu.edu.sg/home/ehchua/programming/sql/Relational_Database_Design.html[fo rrelationaldatabasedesign]
3. http://docs.oracle.com/cd/A97335_02/apps.102/a81358/05_dev1.htm[ForER Diagram]
4.http://plsql-tutorial.com/[ForPL/SQL]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Understand basic concepts regarding data, database systems and various data models.

C02 : Understand various data modeling techniques and entity relationship models.

C03 : Learn about relational data model and related concepts.

C04 : Gain insights into database design and normalization concepts.

C05: Understand various database terminologies and map it in application.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to database systems	✓				
2	Data modeling using entity relationship models		✓			
3	Relational data modeling			✓		
4	Database design methodology				✓	

5	Schema definition, Constraints and queries					✓
6	Advanced query processing					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	-	2	-	3	-	-	-
CO2	2	2	-	2	2	2	-	-
CO3	3	3	3	3	-	2	-	-
CO4	-	2	-	3	-	2	-	-
CO5	-	-	2	3	2	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA409: Computer Oriented Numerical Methods

(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theory lectures illustrations of certain real world problems requiring numerical computations inference will be explained. Numerical methods will be introduced as tools for solving such problems also be emphasized.

Outline of the Course:

Unit No.	Title of the Unit	Minimum number of hours
1	Numerical Methods	05
2	Interpolation	07
3	Numerical Integration	06
4	Solution of Linear Equation	08
5	Solution of Ordinary Differential Equations	06
6	Quantitative Techniques	04

Total Hours (Theory): 36
Total: 36

Detailed Syllabus:

Unit-I: Numerical Methods Hours: 05
Numerical Methods: Bisection, False-Position, Newton-Raphson methods

Unit-II: Interpolation Hours: 07
Interpolation: Difference tables, Newton forward and backward Interpolation formula, Lagranges formula, Newton's Divided Difference Formula

Unit-III: Numerical Integration Hours: 06
Numerical Integration: Trapezoidal rule, Simpson’s 1/3 & 3/8 rules

Unit-IV: Solution of Linear Equation Hours: 08

Solution of Linear Equation: Gauss-Jordan Elimination methods, Gauss - Seidal and Gauss Jacobi Interactive methods

Unit-V: Solution of Ordinary Differential Equations Hours: 06

Solution of Ordinary Differential Equations:Taylor Series and Euler Methods, Runge-Kutta Methods,

Unit–VI: Quantitative Technique Hours: 04

Linear Programming Problem - Formulation, Solution by Graphical Method.

Core Books:

1. D. N. Datta: Computer Oriented Numerical Methods, Vikas Pub. House, 2008.
2. S.S. Sastry : Introductory methods of Numerical Analysis, Prentice Hall of India
3. J.K. Sharma:Operation Research, Macmillan India Limited

Reference Books:

1. V. Rajaraman Computer Oriented Numerical Methods, 3rd Edition, Prentice-Hall of India Pvt.Ltd
2. Miller :Mathematical Statistics with applications, Pearson
3. E. Balaguruswamy: Numerical Methods, TMH

Web References:

1. https://ece.uwaterloo.ca/~dwharder/NumericalAnalysis/10RootFinding/bisection/examples.html [Bisection Method]
2. https://ece.uwaterloo.ca/~dwharder/NumericalAnalysis/10RootFinding/falseposition/examples.html [False Position Method]
3. file:///C:/Users/f1cmpica-1/Downloads/Numerical%20Analysis%20-%20MTH603%20Handouts%20Lecture%20%2021.pdf [Interpolation Newton's Forward Difference Formula]
4. https://paws.kettering.edu/~ktebeest/math305/simp38b.pdf[Simpson's–1/3 & Simpson's–3/8]

Course Outcomes: Upon successful completion of the course, the students will:

C01: Understand practical use of numerical methods in the field of computer science and applications and applying iterative methods to find out approximate root of nonlinear equation.

C02: Able to estimate the missing data through various interpolation methods.

C03: Familiarize with numerical integration and finding approximate solution of a definite integral.

C04: Solve systems of linear equations using various methods including Gaussian and GaussJordan elimination and inverse matrices. Able to identify when to apply direct and indirect method to solve linear equation.

C05: Compute numerical solution of an ordinary differential equations using methods like Taylor Series and Euler Methods, Runge-Kutta methods.

C06: Introduce how operation Research approach helps decision makers in making rational and effective decisions and formulating and solving LPP using Graphical method. Also how to write dual form of LPP.

Course Outcomes Mapping :

Unit No.	**Unit Name**	**Course Outcomes**					
		C01	**C02**	**C03**	**C04**	**C05**	**C06**
1	Numerical Methods	✓					
2	Interpolation		✓				
3	Numerical Integration			✓			
4	Solution of Linear Equation				✓		
5	Solution of Ordinary Differential Equations					✓	
6	Quantitative Techniques						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	1	2	1	1	3	1	1	2
CO2	1	2	-	-	2	-	-	1
CO3	1	1	1	1	3	-	-	1
CO4	1	1	1	-	1	-	-	2
CO5	1	1	1	-	1	-	-	1
CO6	1	3	3	1	3	1	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA410: Fundamentals of Commerce and Business Processes

(100 Marks)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory sessions, the basic terminology along with concrete Illustrations will be discussed. The fundamentals in relation with different industries will be taught and demonstrated so that students can visualize the various business entities and commercial processes.

Outline of the Course:

Unit No.	Title of the Unit	Minimum number of hours
		Theory
1	Basics of Accounts	08
2	Cost Accounting	04
3	Inventory Valuation	07
4	Financial Management and Business Organization	06
5	Banking	05
6	Human Resource & Marketing Management	06

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit - I: Basics of Accounts Hours: 08

Introducing the concepts of Accounting principles and convictions, objectives of Financial Accounting, Double Entry Book-keeping System, recording financial transaction, preparation of Statutory Financial Statements, Profit & Loss account: (Balance sheet, Cash flow Statement), analysis of Financial Statement: (Ratio Analysis, Common size B/S, Comparative Financial Statement).

Unit - II: Cost Accounting Hours: 04

Meaning of Cost, Costing and Cost Accounting, Methods of Costing, Marginal Costing, Breakeven Analysis, Make or buy decision.

Unit - III: Inventory Valuation Hours: 07

Introduction of inventory, Methods of pricing, FIFO, LIFO, Weighted average, Treatment of returns from previous issues and shortages identified at stores verification.

Unit - IV: Financial Management and Business Organization   Hours: 06

Meaning and functions of Financial Management, sourcing of funds and cost of funds, Optimum utilizations of funds, Time value of money, Investment decision, Dividend decision, Working Capital Management, Cash flow, Taxation. Forms of business organization, sole proprietorship firm, Partnership firm, Joint stock Company, Cooperative Society.

Unit - V: Banking   Hours: 05

Introduction to Banking, Deposits Management, Types and Schemes, Management of Advances, Types of Management, Banking Services, Banking Products.

Unit - VI: Human Resource & Marketing Management   Hours: 06

Human Resource: HR planning in context of software industries, Selection, Development and Training, Welfare and working conditions.

Marketing Management: Selling and Marketing comparison, Core concepts of marketing, Pricing Decisions, Advertising and Sales promotion.

Core Books:

1. Dr. S.N. Maheshwari: Financial Accounting, Sultan Chand Publication.
2. I.M.Pande: Financial Management.
3. V.S.P. Rao: Human Resource Management.

Reference Books:

1.Dr. S.N. Maheshwari: Advanced Accounting, Sultan Chand Publication.

2.M.N. Arora: Advanced Cost Accounting, Himalaya Publishing House

3. B.S.Mathur: Banking Law & Practice.
4. Prof D R Patel Accounting and financial management.

Web References:

1. http://iws.collin.edu/ost/pdfs/ACNT1303/ACNT1303lecture_notes.pdf [Introduction of Accounting]
2. http://www.ucosbdc.org/Websites/ucosbdc/Images/accounting_basics.pdf
   [Accounting Equation, Balance Sheet]
3. http://wiki.svtuition.org/2009/03/basic-accounting-notes.html [Area of Accounting, Accounting Elements]
4. Training and Development Practices In Indian Hotel Industry: An Empirical Investigation
   International Journal of Marketing and Human Resource Management

(IJMHRM), Volume 1 * Issue 1 * May 2010, http://www.iaeme.com/MasterAdmin/UploadFolder/paper4.pdf [Human Resources]

5. Making Better Marketing Decisions Faster with Accounting Data http://www.omicsgroup.org/journals/making-better-marketing-decisions-faster-with-accounting-data-2168-9601.1000e116.pdf [Marketing Management]
6. How small businesses master the art of competition through superior competitive advantage, http://www.aabri.com/manuscripts/121156.pdf[Marketing Management]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Be able to understand areas of accounting and learn Financial Accounting in detail.

C02 : Be familiar with Costing and understand Break-even analysis, and determine cost using different methods and techniques.

C03 : Be able to understand periodical and perpetual inventory and apply inventory calculation methods.

C04 : Be able to understand the Financial Management and Business Organization.

C05 : Be able to understand different types of sectors and their processes.

Course Outcomes Mapping:

Unit No.	Unit Name		Course Outcomes			
		C01	**C02**	**C03**	**C04**	**C05**
1	Basics of Accounts	✓				
2	Cost Accounting		✓			
3	Inventory Valuation			✓		
4	Financial Management and Business Organization				✓	
5	Banking					✓
6	Human Resource & Marketing Management					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	-	3	3	-	-	-	-
CO2	-	2	3	3	-	-	-	-
CO3	-	-	3	3	-	-	-	-
CO4	-	-	3	3	-	-	-	-
CO5	-	-	-	-	2	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS126.01 (C):
Communication Skills – I

I. **Credits and Schemes**

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HS126.01 A/B/C/D/E/F/G/H	Communication Skills – I	02	02	30	70	--	--	100

II. **Course Outline**

Module No.	Title/Topic	Classroom Contact Sessions
1	**Communicative English** • Time and Tenses • Active and Passive Voice • *Direct-Indirect Speeches* • Prepositions and Conditionals	**04**
2	Functional English • Introduction to Functional English • *Describing Actions and Processes, Offering, Requests, Routines/Timetable, Making Comparisons, Sharing Interests and Experience*	**04**
3	Conversational Skills • *An Introduction to Conversations for various purposes* • Importance of acquiring Conversational Skills • Models, Techniques and Types of Conversations	**04**
4	Introduction to Communication and Key Concepts in Communication • An Introduction to Communication • *Basic Terms, Concepts, and Contexts of Communication*	**06**

	• *Factors influencing message encoding, the nature of message, and message uses and effects* • *Importance, Types and Principles of Communication*	
5	Effective Listening and Reading Skills • An Introduction to Listening and Reading • *Purposes, Types and Techniques of Listening and Reading* • *Barriers to effective Listening & Reading and overcoming the Barriers* • *Note-taking and Note-making*	**06**
6	Writing Skills • An Introduction to Writing • Importance of effective Writing • *Paragraph Development: Coherence – Topic Sentence, Supporting Sentence & Data etc.* • Business Letter Writing	**06**
	Total	**30**

III. Instruction Methods and Pedagogy

The course is based on pragmatic learning. Classroom Teaching will be facilitated by Reading Material, Classroom Discussions, Task-based learning, projects, assignments and various interpersonal activities like case-studies, critical reading, group-work/pair-work, and presentations.

IV. Evaluation:

Students will be evaluated continuously in the form of internal as well as external examinations. Evaluation (Theory) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment / Project Work	2	25	25
2	Attendance and Class Participation			05
			Total	**30**

External Evaluation

The University Theory examination will be of 70 marks and will test the reasoning, logic and critical thinking skills of the students by asking them theoretical as well as application based questions. The examination will avoid, as far as possible, grammatical errors and will focus on applications. There will be at least one question on case analysis relevant to the components of the course.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Theory Paper	01	70	70
			Total	70

Course Outcome (COs):

After completion of the course, the student would :

CO1	should have polished their grammar and developed the ability to communicate effectively in business situations,
CO2	be able to communicate message accurately, handle situation that require thoughtful communication, to use appropriate words and tones and so on.
CO3	be able to use the language effectively for various functions
CO4	develop familiarity with varied styles of communication and gain insights into how to deal with people with different communication styles
CO5	hone basic linguistic and communication skills (of students) required in a business organization, namely: Listening, Speaking, Reading and Writing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	3	-	2
CO5	-	-	-	-	-	3	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books:-

- Sanjay Kumar and PushpLata (First Edition, 2011), *Communication Skills*, Oxford University Press, New Delhi
- Krishna Mohan and MeenaBanerji (2010), *Developing Communication Skills*, Macmillan Publications India Ltd., New Delhi

- M V Rodriques (2013), *Effective Business Communication*, Concept Publishing Company (P) Ltd., New Delhi
- Mohan and Meenakshi Raman (2006), *Effective English Communication Krishna,*Mcgraw-Hill Publishing Company Limited, New Delhi
- Geoffrey Leech & Jan Swartvik (1994), *A Communicative Grammar of English,* Longman Publications, New York
- Jones Leo (1979), *Functions of English*, Cambridge University Press, UK

Reference Reading:-

- http://www.communicationskills.co.in/index.html
- http://www.hodu.com/default.htm
- http://www.bbc.co.uk/worldservice/learningenglish
- http://www.englishlearner.com/tests/test.html
- http://www.englishclub.com/vocabulary/idioms-body.htm
- http://dictionary.cambridge.org

DETAIL SYLLABUS OF THIRD SEMESTER OF B.Sc.(IT)

Teaching and Examination Scheme (BSc.(IT) Programme)

Effective from Year 2019-20

Semester-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA501	Programming the Internet	4	3	7	7	10	20	70	15	15	70	200
CA502	Fundamentals of Computer Networks	3	-	3	3	10	20	70	-	-	-	100
CA503	Object Oriented Programming Using C++	4	3	7	7	10	20	70	15	15	70	200
CA504	E-Commerce	3	-	3	3	10	20	70	-		-	100
CA505	System Analysis and Design	3	-	3	3	10	20	70	-		-	100
HS122 C	Values and Ethics	2	-	2	2	30		70	-		-	100
	University Elective-I **	-	2	2	2	-		-	30		70	100
		19	08	27	27	600			300			900

** CMPICA has decided to offer **CA 224 Introduction to Web Designing** course for the students of other institutions. CMPICA students will opt any university elective offered by different institutions of the university.

University Electives

No	Course Code	Course Name	Department/Faculty
1	CE281.01	Art of Programming	Engineering
2	CL281.01	Environmental Sustainability & Climate Change	Engineering
3	EE283	Python for Electrical Engineering	Engineering
4	IT281.01	ICT Resources & Multimedia	Engineering
5	ME281.01	Engineering Drawing	Engineering
6	PH233.01	Fundamentals of Packaging	Pharmacy
7	PD260.01	Basic Laboratory Techniques	Applied Science
8	NR251.01	First Aid & Life Support	Nursing
9	PT191.01	Health Promotion & Fitness	Physiotherapy
10	BM231	Banking & Insurance	Management
11	EC281.01	Introduction to MATLAB Programming	Engineering
12	PD261	Astrophysics, Space and Cosmos-I	Applied Science

CA501: Programming the Internet
(Marks 200)

Contact Hours: 07

Objective: The objective of the course is to make students familiar with design and develop process of a website using different elements of HTML and link pages to create a website. Students should be able to design and develop dynamic and professional web pages such as e-commerce and academic contents with client side scripting and active management.

Pre- requisite: None.

Methodology & Pedagogy: During theory sessions, topics related to web designing technologies will be covered with suitable examples. During Practical sessions, students will be required to design and develop entire web sites using several web designing technologies and editors.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Basics in Web Design	06	36
2	Web Page Designing using HTML5	10	
3	Introduction to Cascading Style Sheets	10	
4	JavaScript Programming I	08	
5	JavaScript Programming II	07	
6	HTML Template and Web Publishing	07	

Total Hours (Theory) : 48
Total Hours (Lab) : 36
Total : 84

Detailed Syllabus:

Unit I: Basics in Web Design Hours: 06

Introduction to WWW, URL, webpage, web site, web servers, web browser, Web Application, Client and Server side scripting languages, types of websites, Web Design and Development, Different phases of a web project.

Unit II: Web Page Designing using HTML5 Hours: 10

An introduction to HTML, Basic Structure of an HTML Document, HTML tags, Text and Paragraph formatting tags and its attributes, Lists, table tags and its attributes, Hyperlinks, Inserting an Image in webpage, Form tags and its attributes
Advance Elements of HTML: ! Doctype, meta, Semantic element, New Input type elements, Multimedia tags

Unit II: Introduction to Cascading Style Sheets Hours: 10

Introduction to Cascading Style Sheet (CSS), basic syntax and structure, CSS selectors, Ways of specifying style, CSS Properties, CSS Styling (Background, Text, Fonts, Lists, Tables, Links) , CSS Box Model , CSS Navigation Bar

CSS3:- CSS Rounded Corners, Box & Text Shadow , Gradients, Background Images, Transitions, Transforms, and Animations

Unit IV: JavaScript Programming_I Hours: 08

Introduction to Client Side Scripting, Basics of Java Script, Java Script Statements, Comments, variables, Operators and Expressions, Conditions statement, Functions, Dialog boxes.

Unit V: JavaScript Programming II Hours: 07

The Java Document Object Model (DOM): Introduction, JSSS DOM, Objects in HTML, Object hierarchy, Event handling.

Forms: Form object, built in objects, User defined objects, Cookies, Java Script Window
DHTML: Introduction to DHTML, DHTML CSS, DHTML Java Script, DHTML HTML DOM, DHTML Events.

Unit VI: HTML Template and Web Publishing Hours: 07

Creating and Saving the site using Sublimetext, HTML template creation and Integrating with other pages, FTP Management: Understanding FTP, Setting up FTP Server, Uploading and downloading FTP contents

Core Books:

1. Matthew MacDonald: HTML5 The Missing Manual, O'Reilly Media, August 2011.
2. Peter Gasston: The Book of CSS3, A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.

3. Richard York: Beginning CSS Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2005.

Reference Books:

1. Ivan Bayros : Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, fourth revised edition , BPB Publication.
2. David Mc Farland : CSS The Missing Manual, O'Reilly, 2006.
3. Julie C. Meloni : HTML, CSS and JavaScript All in One, Pearson.

Web References:

1. http://spyrestudios.com/freecontentmanagementsystems/[For Introduction to Content Management System]
2. http://www.tutorialspoint.com/html5 [For notes on HTML5 tags]
3. http://www.w3schools.com/html/ [For HTML5, CSS and JavaScript notes and examples]
4. https://in.godaddy.com/help/dreamweaver-cs6-publish-your-website-7811[Publish your website using Dreamweaver]
5. http://fullbooksfreedownload.blogspot.in/2016/02/html-css-javascript-web-publishing-in.html [Book :- HTML, CSS & JavaScript Web Publishing in One Hour a Day, Sams Teach Yourself, 7th Edition PDF]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Get familiar with basics of the Internet Programming.

C02 : Gain deep understanding of the implementation of HTML tags and able to create HTML static web pages.

C03: Able to use basic properties of CSS and CSS3 in html file

C04 : Able to design template using HTML5 and CSS3

C05 : Able to create dynamic web page using JavaScript

C06: Students will apply their knowledge to create interactive websites and template using sublimetext editor.

C07: Able to understand and set up FTP Server.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes						
		C01	C02	C03	C04	C05	C06	C07
1	Basics in Web Design	✓						
2	Web Page Designing using HTML5		✓		✓		✓	
3	Introduction to Cascading Style Sheets			✓	✓		✓	
4	JavaScript Programming I					✓	✓	
5	JavaScript Programming II					✓	✓	
6	HTML Template and Web Publishing						✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	-	-	-	-	-	-	-
CO2	1	3	-	-	-	-	-	-
CO3	-	-	1	-	-	-	-	-
CO4	-	-	2	2	2	2	-	3
CO5	-	-	3	2	2	-	-	-
CO6	-	-	2	2	-	2	-	-
CO7	1	-	-	-	3	2	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA502: Fundamentals of Computer Networks

(Marks 100)

Contact Hours: 03

Pre-requisite: None

Methodology and Pedagogy: Theory lectures will be focused on the basic concepts of computer networks and terminologies used in computer networking. The teacher will also discuss various network reference models and the functionality of the layers of different models. The teacher will discuss variety of design issues related to layers of model, networking devices and networking protocols.

Outline of the course:

Unit No.	Title of Unit	Minimum Number of Hours
		Theory
1	Basics of Computer Network	05
2	Network Communication Concepts	07
3	Reference Models	05
4	Data Link Layer Design	08
5	Network Layer Design	06
6	Network Protocols & Devices	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Basics of Computer Network Hours: 05

What is Computer Network, Aim and objective of Computer Network: Applications of Computer Network, Merits & Demerits of Computer Network, Examples of Computer Network, Components of Computer Network, Types of Networks, Network Topologies.

Unit -2:Network Communication Concepts Hours: 07

Parallel & Serial Communication, Synchronous & Asynchronous Communication.

Modes of Communication: Simple, Half-Duplex, Full-Duplex, Multiplexing, De-Multiplexing: FDA, TDM. Communication Media: Guided Media, Unguided Media. Switching Techniques: Circuit Switching, Message Switching, Packet Switching.

Unit -3: Reference Models Hours: 05

OSI Reference Model: Architecture, Function of Each Layer, TCP/IP Reference Model: Architecture, Function of Layers. Comparison of OSI & TCP/IP Models.

Unit -4: Data Link Layer Design Hours: 08

Services provided to the Network Layer, Framing, Error Control (error detection and correction code), Flow Control, Data Link Layer in the Internet (SLIP, PPP). MAC sub layer: CSMA/CD/CA, IEEE standards (IEEE802.3 Ethernet IEEE 802.4 Token Bus, IEEE 802.5 Token Ring).

Unit -5: Network Layer Design Hours: 06

Routing Algorithms, Congestion Control Policies, Concept of Internetworking.

Unit -6: Network Protocols & Devices Hours: 05

Protocol Hierarchy, Network Protocols: SMTP, FTP, MIME, POP, Telnet, DNS, DHCP, ARP, RARP. Network Devices: Hub, Switch, Repeater, Router, Gateway, Bridge.

Core Books:

1. Andrew S. Tanenbaum: Computer networks, 4th Edition, Pearson Publication.
2. Behrouz Forouzan: Data communications and networking, 4th Edition, McGraw Hill.

Reference Books:

1. Achyut S Godbole, AtulKahate: Data Communications and Networks, 2nd Edition, McGraw Hill.
2. Honey Cutt: Using the Internet, 4th Edition, PHI Learning.

Web References:

1. http://nptel.ac.in/ [Online Lectures].
2. http://www.tutorialspoint.com/data_communication_computer_network/index.html [Course Material, Notes].

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Understand basic of networks, networking technologies and concepts.

C02 : Come to know how communication systems work.

C03 : Learn different networking reference models

C04 : Gain understanding of various layers' design issues and their solutions

C05 : Learn functionality of various network protocols and how devices work

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Basics of Computer Network	✓				
2	Network Communication Concepts		✓			
3	Reference Models			✓		
4	Data Link Layer Design				✓	
5	Network Layer Design				✓	
6	Network Protocols & Devices					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	-	-	2	-	-	-	2
CO2	2	2	-	-	2	-	2	-
CO3	-	-	2	3	-	2	-	-
CO4	3	3	2	-	2	-	1	2
CO5	2	-	-	-	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA503: Object Oriented Programming using C++

(Marks 200)

Contact Hours: 07

Pre- requisite: None.

Methodology & Pedagogy: During theory lectures, illustrations of certain complex real world applications, which emphasis the use of advanced concepts of programming, will be discussed. The fundamental concepts regarding development activities, various object oriented systems and other advanced issues in object oriented programming will also be discussed. In addition, there may be announced or unannounced quizzes/assignments. During Practical sessions, students will be required to carry out case studies using the concepts and techniques they have learnt during theory sessions.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Foundation of C++	06	36
2	Classes and Objects	10	
3	Operator Overloading and Type Conversion	10	
4	Inheritance, Virtual Functions and Polymorphism	09	
5	Exception Handling	07	
6	Using Files of IO	06	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit - I: Foundation of C++   Hours: 06
Overview of C-Expressions, Arrays, Functions, Structures, Pointers- Call by value, call by reference, Introduction to C++, Structure of C++ program, Fundamental of OOP, Characteristics of OOP, inline function, Default arguments, function overloading

Unit - II: Class and Objects   Hours: 10

Define class, objects, visibility modes, static members, friend function, Constructors and destructors, Default Constructor, Copy constructors, Parameterized Constructor.

Unit - III: Operator Overloading and Type ConversionHours: 10

Introduction, Operator overloading through member function, Need of Function objects- user defined Conversions.

Unit - IV: Inheritance, Virtual Functions and PolymorphismHours: 09

Introduction, Base class access controls, Inheritance and protected members, Virtual base classes, Virtual functions – calling through a base class, pure virtual functions, abstract class, early vs. late binding

Unit - V: Exception Handling Hours: 07

Introduction, Exception handling fundamentals, catching class types, using multiple catch statements, exception handling options, handling derived class exceptions, applying exception handling

Unit - VI: Using Files for IO Hours: 06

<fstream> and the file class, opening and closing file, reading and writing text files, Unformatted and Binary IO, more get() functions, detecting EOF, Random access, I/O status

Core Books:

1. Herbert Schildt: C++ A Beginner's Guide, TMH Publications.
2. E Balagurusamy: Object Oriented Programming with C++, 2nd Edition, TMH Publications.
3. Sourav Sahay: Object Oriented Programming with C++, 2nd Edition, Oxford University Press
4. Bhushan Trivedi: Programming with ANSI C++, 2nd Edition, Oxford University Press
5. Bjarne Stroustrup: The C++ Programming Language, 3rd Edition, Pearson Education

Reference Books:

1. Robert Lafore: Object Oriented Programming in C++, 4th Edition, Sams Publications.
2. YashavantKanetkar: Let Us C++, 8th Edition, BPB Publications.
3. Herbert Schildt: The Complete Reference C++, 3rd Edition, TMH Publications.

Web References:

1. http://www.cplusplus.com/doc/tutorial/[For Entire Syllabus]
2. http://www.learncpp.com/[For coding standards and concepts]
3. http://www.tutorialspoint.com/cplusplus/cpp_basic_input_output.htm[For Practicals]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : The present course builds up on the knowledge of C programming and basic data concept (array,Arrays, Functions, Structures, Pointers- Call by value, call by reference).

C02 : Understand real problem with the Class and object concepts of C++.

C03 : Learning operator overloading and conversion of built-in and user defined data types.

C04 : To gain understanding of inheritance, the main concept of Object oriented language.

C05 : To learn file handling in Object oriented C++ Language.

Course Outcomes Mapping :

Unit No.	**Unit Name**	**Course Outcomes**				
		C01	**C02**	**C03**	**C04**	**C05**
1	Foundation of C++	✓				
2	Class and Objects	✓				
3	Operator Overloading and Type Conversion		✓			
4	Inheritance, Virtual Functions and Polymorphism			✓		
5	Exception Handling				✓	
6	Using Files for IO					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	-	-	2	-	-	-	2
CO2	2	2	-	-	2	-	3	-
CO3	-	-	2	3	-	2	-	-
CO4	3	3	2	-	2	-	2	2
CO5	2	-	-	2	-	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA504: E-Commerce

(100 Marks)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures, the fundamental concepts regarding Ecommerce, Payment, E-Security, E-branding and E-Data Interchange will be discussed .Students will be required to carry out case studies using the concepts and techniques they have learnt during theory sessions.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours
		Theory
1	Introduction to E-commerce	07
2	E- Marketing	05
3	E-Security and E-Payment	06
4	Internet Bandwidth and Technology Issue	05
5	Mobile Commerce	07
6	E-Commerce Solutions	06

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit - I: Introduction to E-commerce Hours: 07

Emergence and use of Internet, origin of web and world wide web, advantages and disadvantages of Ecommerce, Features of E-Commerce, Application of E-Commerce, Electronic Commerce over the Internet, Types of E-commerce.

Unit - II: E- Marketing: 05

Traditional Marketing, Online Marketing, E- advertising, Internet Marketing Trends and strategies, Ebranding.

Unit - III: E-Security and E-Payment: 06.

Security on the Internet, E-Business risk management issues, Digital token based E-Payment System, Properties of Electronic Cash, Digital Signature.

Unit - IV: Mobile Commerce: 05

Definition of M-Commerce, Features of M-Commerce, Advantages and Disadvantages of MCommerce, Areas of M-Commerce Applications, Payment Systems and Models in M-Commerce.

Unit V: E- CRM and ERP Hours: 07
Customer Relationship Management, CRM capabilities and Customer life cycle, Introduction to ERP, Reasons for the growth of the ERP Market, Advantages and Disadvantages of ERP.

Unit VI: E-Commerce Solutions: 06
Categories of E-Commerce Solutions, Solutions in Detail: Magento, Prestashop, Drupal Commerce, Demandware, Intershop, OXID eShop, RBS Change and Hybris.

Core Books:

1. P.T.Joseph, S.J.: E-Commerce- An Indian Perspective, 3rd Edition, PHI learning Private Limited.
2. Kamlesh K. Bajaj, Debjani Nag: E-Commerce, The Cutting Edge of Business, 2nd Edition, McGraw-Hill Education.

Reference Books:

1. Anita Rosen: The e-commerce Question and Answer Book: A Survival Guide for Business Managers, 2nd Edition, Amacom.
2. Janice Reynolds: The Complete E-Commerce Book: Design, Build & Maintain a Successful Web-based Business, 2nd Edition, CRC Press.
3. By Philippe Humeau&MatthieuJung : The White Book of Ecommerce Solutions, NBS Systems
4. IshitaLahiri and Sujit Kumar Ghose : Principals of Marketing and E-Commerce

Web References:

1. http://www.ecommercetutorial.net/ [For Tutorial]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : To introduce the concept of Electronic Commerce

C02 : To understand the concept of E-marketing.

C03 : To understand the role of E-Security & E-Payment.

C04 : To understand M-Commerce Payment System.

C05 : To learn CRM, ERP and E-commerce Solution.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to E-commerce	✓				
2	E- Marketing		✓			
3	E-Security and E-Payment		✓	✓		
4	Mobile Commerce:			✓	✓	
5	CRM and ERP					✓
6	E-Commerce Solutions					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	2	3	3	3
CO2	3	3	2	2	2	3	3	2
CO3	3	2	3	2	2	3	2	3
CO4	3	2	3	2	2	3	2	3
CO5	3	2	3	2	2	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA505: System Analysis and Design

(100 Marks)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures emphasis will be given on understanding of analyzing business needs, perform planning and analysis of business system, give examples of how Information
Systems are designed and constructed and how major modifications are made to the existing systems.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours Theory
1.	Basics of Information Systems	05
2.	Information Systems Analysis Overview	06
3.	Information Gathering and System Requirement specification	07
4.	Data flow diagrams and Process Specification	07
5.	Overview to Data input and designing output methods	06
6.	Implementation and maintenance of information systems	05

Total Hours (Theory): 36
Total: 36

Detailed Syllabus:

Unit - I: Basics of Information Systems Hours: 05

Types of Information, Understand the concept of System, Need for a Computer-Based Information System, Management Structure, Management and Information Requirements, Qualities of Information, Examples of Information Systems, Various Functions in Organizations, Varieties of Information Systems.

Unit - II: Information Systems Analysis Overview Hours: 06

Overview of Design of an Information System, System Development Life Cycle (SDLC), The Role and Tasks of Systems Analyst, Attributes of a Systems Analyst, Tools Used by Systems Analyst, Approaches to Systems Development.

Unit - III: Information Gathering and System Requirement specification Hours: 07

Strategy to Gather Information, Information Sources, Methods of Searching for Information, Interviewing Technique, Questionnaires, Other Methods of Information Search, System Requirements Specification, Data Requirements, steps in Systems Analysis, Modularizing Requirements Specifications, deciding on Project Goals, Examining Alternative Solutions, Evaluating Proposed Solution, Cost-benefit Analysis, Payback Period, Feasibility Report, System Proposal.

Unit - IV: Data flow diagrams and Process Specification Hours: 07

Symbols Used in DFDs, describing a System with a DFD, Levelling of DFDs, Levelling Rules,
Logical and Physical DFDs, Case Tools to Draw DFD, Process Specification Methods, Structured English, Decision
Table Terminology and Development, Extended Entry Decision Tables, Eliminating Redundant Specifications

Unit - V: Overview to Data input and designing output methods Hours: 06

Data Input, Coding techniques, validating input data, interactive data input, objective of output designing, design of screens and output reports.

Unit - VI: Implementation and maintenance of information systems Hours: 05

Control in Information Systems, The Objectives of Control, Control Techniques, Audit of Information Systems, Security of Information Systems, designing reliable and maintainable systems, design of software, software design and documentation tools, managing quality assurance, user training, training methods and conversion.

Core Books:

1. James A Senn: Analysis and Design of Information System, McGraw Hill International, 2003.
2. V. Rajaraman : Analysis and Design of Information Systems, Prentice-Hall of India Private Limited, 2003.
3. Kendall and Kendall: Systems analysis and Design, 5^{th} Edition, Prentice-Hall of India Private Limited, 2003.

Reference Books:

1. Jeffrey L. Whitten, Lonnie D. Bentely and Kevin C. Dittman : Systems Analysis and Design Methods, Tata McGraw Hill Publishing Co. Ltd., 2001.
2. Tuthill and Leavy : Knowledge Based Systems : Mangers Perspectives : Tab professional and Reference Books, 1991.

Web references:

1. http://lecture-notes-forstudents.blogspot.in/2010/04/system-analysis-and-designsad.html[For referring SDLC]
2. http://www.slideshare.net/aroravinay/1-introduction-to-ado-infosys[For Introduction of System analysis and design]

3. http://bcastuff.blogspot.in/p/sad-notes.html#.U6uW-ZzWIeo[For brief notes on System analysis]
4. http://www.eis.mdx.ac.uk/staffpages/geetha/bis2030/DFD.html[For DFD notes and examples] 5. http://faculty.washington.edu/ytan/is460/notes/LN11.pdf[For DFD notes and examples]

Course Outcomes: Upon successful completion of the course, students will:

C01 : Understand basic concepts of system, types of information, qualities of information and its flow with respect to management structure

C02 : Gain knowledge about how the system will develop and guide by well-experienced person in the organization with its documentation.

C03 : Learn about various tools used to design flows of data, processes and data stores, performed by users of the system in graphical way.

C04 : Understand various designing methods used to solve a problem while taking input and producing output as per given specification.

C05 : Gain knowledge about various concepts of system maintenance and conversions

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Basics of Information systems	✓				
2	Information system analysis Overview		✓			
3	Information gathering and System Requirement Specification		✓			
4	Data Flow Diagram and Process specifications			✓		
5	Overview to data input and design output methods				✓	
6	Implementation and maintenance of information systems					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	3	3	2	3	2	2
CO2	3	2	2	3	3	2	2	2
CO3	3	2	2	2	2	2	2	3
CO4	3	2	3	3	2	3	2	2
CO5	3	2	2	2	3	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS122C:
VALUES AND ETHICS (Sem-III)

I. **Credits and Schemes:**

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
III	HS122 C	VALUES AND ETHICS	02	02	30	70	-	-	100

II. **Course Outline**

Module No.	Title/Topic	Classroom Contact Hours
1	**Introduction to Values and Ethics** • *Need, Relevance and Significance of Values and Ethics : General* • *Concept and Meaning of Values and Ethics*	06
2	**Elements and Principles of Values** • *Universal & Personal Values* • *Social, Civic & Democratic Values* • *Adaptation Models & Methods of Values*	08
3	**Applied Ethics** • *Universal Code of Ethics* • *Professional Ethics* • *Organizational Ethics* • *Ethical Leadership* • *Domain Specific Ethics*	08
4	**Value, Ethics & Global Issues** • *Cross-Cultural Issues* • *Role of Ethics & Values in Sustainability* • *Case Studies*	08
	Total	30

III. **Instruction Methods and Pedagogy**

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, critical reading, group work, independent and collaborative research, presentations etc.

IV. **Evaluation:**

The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (Theory) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
3	Assignment / Project Work	2	25	25
4	Attendance and Class Participation			05
			Total	30

External Evaluation

The University Theory examination will be of 70 marks and will test the reasoning, logic and critical thinking skills of the students by asking them theoretical as well as application based questions. The examination will avoid, as far as possible, grammatical errors and will focus on applications. There will be at least one question on case analysis relevant to the components of the course.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Theory Paper	01	70	70
			Total	70

Course Outcome (COs):

After completion of the course, the student would :

CO1 Understand the concepts and mechanics of values and ethics.

CO2 Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.

CO3 Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.

CO4 Develop their responsibility towards society.

CO5 Comprehend their own core values and adhere to those values at their workplace.

CO6 Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	3	-	-
CO4	-	-	-	2	-	3	-	-
CO5	-	-	-	-	-	3	-	2
CO6	-	-	-	-	-	3	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Reference Books / Reading

- Human Values and Ethics in Workplace, United Nations Settlement Program, 2006. (http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
- Ethics for Everyone, Arthur Dorbin, 2009. (http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf) .
- Values and Ethics for 21st Century, BBVA. (https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf) www.ethics.org

DETAIL SYLLABUS OF FOURTH SEMESTER OF B.Sc.(IT)

Teaching and Examination Scheme (BSc.(IT) Programme)
Effective from Year 2019-20
Semester-IV

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		rnal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA506	Data Structures & Algorithms	4	3	7	7	10	20	70	15	15	70	200
CA507	Advanced Database System	-	3	3	3	-	-	-	15	15	70	100
CA508-512	Elective – I	4	3	7	7	10	20	70	15	15	70	200
CA513	Management Information System	3	-	3	3	10	20	70	-		-	100
HS133 C	Creativity, Problem Solving and Innovation	-	2	2	2	-		-	30		70	100
CL142.01	Environmental Sciences	-	2	2	2	-		-	30		70	100
	University Elective-I I**	-	2	2	2	-		-	30		70	100
		11	15	26	26	300			600			900

** CMPICA has decided to offer **CA225 Programming the Internet** course for the students of other institutions. CMPICA students will opt any university elective offered by different institutions of the university.

Elective-I

1.CA508 Graphics Designing
2.CA509 Open Source Technology
3.CA510 Multi Paradigm Scripting
4.CA511 Foundation of Data Science
5.CA512 Fundamentals of Embedded Programming

University Electives

No	Course Code	Course Name	Department/Faculty
1	EC282.01	Prototyping Electronics with Arduino	Engineering
2	CE282.01	Web Designing	Engineering
3	CL282.01	Basics of Environmental Impact Assessment	Engineering
4	EE286	Computer Programming for Electrical Engineering	Engineering
5	IT282.01	Internet Technology & Web Design	Engineering
6	ME282.01	Material Science	Engineering
7	PH238.01	Cosmetics in Daily Life	Pharmacy
8	NR261.01	Life Style Diseases & Management	Nursing
9	PT192.01	Occupational Health & Ergonomics	Physiotherapy
10	BM241	Health Care Management	Management
11	PD262	Astrophysics, Space and Cosmos-II	Applied Science

CA506 : Data Structures and Algorithms

(Marks 200)

Contact Hours: 07

Pre-requisite: Introduction to programming in C, Advanced programming in C and Object oriented programming using C++

Methodology & Pedagogy: During theory lectures emphasis will be given on understanding fundamental concepts of data structures and algorithms and how to use and choose the most efficient data structure from a variety of available ones for different kind of applications. During practical session emphasizes given on designing, analyzing and implementation of all kind of data structure for solving computational problems using high level programming language.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1.	Introduction to Data Structures and Algorithms	06	36
2.	Linear Data Structures Using Array	09	
3.	Dynamic Data Structures	10	
4.	Trees Data Structure	08	
5.	Graph Data Structure	08	
6.	Sorting and Searching	07	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit – I: Introduction to Data Structures and Algorithms Hours: 06

Atomic and composite data, Datatype, Data object, DataStructure, Abstract Datatype, Types of Data Structures, Introduction to Algorithms, Relationship among Data, Data Structures and Algorithms, Analysis of Algorithms, space and time complexity algorithm.

Unit – II: Linear Data Structures Using Array Hours: 09

Linear Data Structures using sequential organization, Concepts of Ordered list, Representation of Stacks using sequential organization, Applications of Stack, Recursion, Application of Recursion, Concepts of Queues, Realization of Queues using sequential organization, Multi- queue, Dequeue, Priority Queue, Applications of Queue.

Unit – III: Dynamic Data Structures Hours: 10

Linked list, Comparison of sequential and linked organizations, Dynamic Memory Management, singly linked list, doubly linked list, Applications of linked list.

Unit – IV: Trees Data Structure Hours: 08

Basic Terminologies, Definition and concepts, Representation of Binary Tree, Operations on Binary Tree and algorithms, Types of Binary Tree, BTree, B+Tree Indexing, AVL trees.

Unit – V: Graph Data Structure Hours: 08

Graph Terminology, Representation of Graph, Operations on Graph, Applications of Graph Structure, minimum spanning tree.

Unit – VI: Sorting and Searching Hours: 07

Sorting Notations and Concepts, Sorting Techniques, Sequential Searching, Binary Searching, Search Trees, Hash Table Methods.

Core Books:

1. Michael T. Goodrich, Roberto Tamassia, David Mount: Data Structures and Algorithms in C++, Wiley; Second edition (2016).
2. R. S. Salaria: Data Structures & Algorithms Using C++, Khanna Book Publishing - 2015.
3. R. Singh, I. Singh: Data Structures: An Algorithmic Approach with C++, Khanna Book Publishing - 2014.

Reference Books:

1. D.S. Malik - Creighton University: Data Structures Using C++, Cengage Learning (2007).
2. Classic Data Structures by Samanta, Debasis.

Introduction to data structures with application-Jean- Paul Tremblay &G.Sorenson

Web References:

1. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/ds_ToC.html[For materials of Data Structures and Algorithms]
2. http://www.cs.usfca.edu/~galles/visualization/[For Data Structures Visualization]
3. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/mst.html

[For Graph Data Structures]

4. http://interactivepython.org/runestone/static/pythonds/Trees/trees.html [For Tree Data Structures]
5. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/niemann/s_man.pdf [For Sorting and Searching Cook book]
6. http://www.cs.princeton.edu/~rs/AlgsDS07/10Hashing.pdf [For Hashing Materials]
7. http://nptel.ac.in/video.php?subjectId=106102064[For Data Structures videos]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Analyze the asymptotic performance of algorithms

C02 : Explore linear and nonlinear data structures

C03 : Apply important algorithmic design paradigms and methods of analysis.

C04 : Use efficient data structures which are a key to designing efficient algorithms.

C05: Compare various sorting and searching techniques

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to Data Structures and Algorithms	✓	✓			
2	Linear Data Structures Using Array		✓	✓		
3	Dynamic Data Structures		✓	✓		
4	Trees Data Structure		✓		✓	
5	Graph Data Structure		✓		✓	
6	Sorting and Searching					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	3	2	2	3	1	1	1
CO2	1	3	3	2	3	1	1	1
CO3	2	3	3	3	3	1	1	1
CO4	2	3	3	3	3	1	1	1
CO5	2	3	3	3	3	1	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA507: Advanced Database System

(100 Marks)

Contact Hours: 03

Pre-requisite: Database Fundamentals.

Methodology & Pedagogy: During the practical session student will learn how to create procedure and function. Students can understand the concept of reusability. Student will learn how to handle error using exception and using trigger they can understand how to restrict the operation at particular time or given condition.

Outline of the Course:

Week No	Practical	Description
1 – 2	Datatypes of PL/SQL, Looping Structures, Conditional Statements, Operators	Basic syntax of PL/SQL Block and usage of Conditional statements and Looping structure inside the block
3 – 5	Implicit Cursor, Explicit Cursor, Parameterized Cursor	Usage of Cursor and its working
6 – 7	User defined Procedure & Function	Student will learn how to create procedure and function and its execution
8 – 9	Error Handling using pre-defined exception and user-defined exception	How to create exception and how to use predefined exception
10 – 11	Row level and Statement level Trigger, GRANT & REVOKE	Syntax of trigger and its working, Assign privileges at different level
12	Import/Export Facility	Student will learn how to import and export database with procedure, function, trigger, tables and data. And to reuse the existing database with other applications.

Total Hours: 36

Core Books:

1. SQL, PL/SQL – The programming Language Oracle - by Ivan Bayross
2. Teach Yourself Database Technologies - by Ivan Bayross

Reference Books:

1. Oracle 9i, PL/SQL Programming by Scoot Urban, Oracle Press
2. Practice book on SQL and PL/SQL by Anjali, Amisha, Roopal and Nirav publications.

Web References:

1. http://plsql-tutorial.com/index.htm[For PL/SQL]
2. http://www.tutorialspoint.com/plsql/plsql_collections.htm [For PL/SQL]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Describe PL/SQL basic datatypes and Code constructs
C02 : Solve database problems using different types of cursors
C03 : Solve database problems using different database objects (stored procedure and user defined function)
C04 : Describe basics of error handling while solving database problems
C05 : Define and describe the control user access to the database
CO6 Describe facilities to re-use the database with other applications

Course Outcomes Mapping:

Week No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1-2	PL/SQL datatypes and control structures	✓					
3-5	PL/SQL data retrieval using cursors	✓	✓				
6-7	Stored procedures and user defined functions			✓			
8-9	Exception Handling	✓	✓		✓		
10-11	Control User Access			✓		✓	
12	Logical backup and re-use facilities of database using Import and Export			✓			✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	1	-	2	-	-	-	-	-
CO2	-	1	3	2	-	-	-	-
CO3	-	3	-	2	-	1	1	-
CO4	-	-	-	3	-	-	2	1
CO5	3	1	2	-	1	-	1	-
CO6	-	-	-	-	2	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA508: Graphics Designing

(100 Marks)

Contact Hours: 07

Pre-requisite: None

Methodology & Pedagogy: During theory lectures foundations of graphics designing related concepts will be introduced to students. Students will be made familiar with the Photoshop and adobe illustrator.

Outline of the Course:

Unit No.	Title of the Unit	Minimum number of hours	
		Theory	Practical
1.	Introduction to Photoshop	08	36
2.	Working with tools in Photoshop	08	
3.	Introduction to Multimedia and Managing Text and Images in Multimedia	09	
4.	Introduction to Adobe Illustrator	07	
5.	Working with Layer and Effect	08	
6.	Graphics & Media Design	08	

Total Hours (Theory): 48
Total Hours (Practical): 36 Total: 84

Detailed Contents:

Unit –I Introduction to Photoshop Hours:08

About Photoshop, navigating Photoshop, menus and panels, opening new files, opening existing files, exploring the toolbox, the new applications bar & the options bar, exploring panels & menus, creating & viewing a new document, customizing the interface, setting preferences, working with images, resizing & cropping images, working with basic selections.

Unit –II Working with tools in Photoshop Hours:08

Getting started with layers, selections tools, painting in Photoshop, photo retouching, introduction to color correction, using quick mask mode, working with the pen tool creating special effects.

Unit – III Introduction to Multimedia and Managing Text and Images in Multimedia Hours:09

Introduction, Usage of Multimedia, Stages of a Multimedia Project, the Multimedia Team. Text Multimedia: Power of Meaning, Fonts and Faces, Using Text in Multimedia, Computers and Text, Font Editing, Hypermedia, Hypertext.

Image in Multimedia: Organizing Tools, Bitmap Images, Vector Drawings, 3-D Drawing and Rendering, Color models, Image File Formats.

Unit – IV Introduction to Adobe Illustrator Hours:07

A Quick Tour of Adobe Illustrator, getting to Know the Work Area, Selecting and Aligning, Creating and

Editing Shapes, Transforming Objects, Drawing with the Pen and Pencil Tools, Color and Painting

Unit – V Working with Layer and Effect Hours:08

Working with Type, working with Layers, working with Perspective Drawing, Blending Colors and Shapes, working with Brushes, Applying Effects, Applying Appearance Attributes and Graphic Styles

• Working with Symbols

Unit – VI Graphics & Media Design Hours:08

Digital painting, Collage making, Designing, Advertisement designing, Brochure designing, Package designing, Preparing artwork for print

Core Books:

1. Tay Vaughan, Multimedia, Making it Work - 7th Edition, Tata McGraw Hill Publication.
2. Lesa Snider, Photoshop CS5 - The missing manual, O'reilly Media, First Edition, May 2010.
3. Lisa DaNaeDayley and Brad Dayley, Photoshop CS5 - Adobe Photoshop CS5 Bible, Wiley India Pvt. Ltd., June 2010.
4. Sharon Steuer ,The Adobe Illustrator CS6 WOW! BooK, Peachpit Press,2013.

Reference Books:

1. Brian Wood , Adobe Illustrator CC, Adobe Press Peachpit
2. Michael Toot, Sherry Kinkoph, Master Adobe Photoshop, Illustrator, Premiere and after Effects Visually, First Edition 2002

Web References:

1.https://www.photoshoptutorials.ws[Photoshop Tutorials]

2.http://www.vectordiary.com/[Illustrator Tutorial]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Able to get familiar with the Photoshop tool, workspace and it's working.

C02 : Able to work with the selection, resize, crop, navigate, color modes, text for editing and creating new images as well as shape.

C03 : Able to understand how Text and Images work in Multimedia

C04 : Use basic Illustrator skills and concepts to develop effective graphics

C05 : Able to create different media designs.

Course Outcomes Mapping:

Unit No.	**Content**	**Course Outcomes**				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to Photoshop	✓				
2	Working with tools in Photoshop		✓			
3	Introduction to Multimedia and Managing Text and Images in Multimedia		✓	✓		
4	Introduction to Adobe Illustrator				✓	
5	Working with Layer and Effect		✓	✓	✓	
6	Graphics & Media Design					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	2	1	2	3	-	-	1
CO2	2	3	1	2	3	-	-	1
CO3	2	2	1	2	3	-	-	1
CO4	2	3	1	2	3	-	-	1
CO5	2	2	1	2	3	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA509: Open Source Technology

(200 Marks)

Contact Hours: 07

Pre-requisite: CA501: Programming the Internet.

Methodology & Pedagogy: During theory lectures, the emphasis will be given on introduction to open source technology, the structure and syntax of PHP, database connectivity using SQL databases, working with forms and user data, etc. During Practical sessions, students will implement the concepts which are discussed in lecture.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1.	Introduction and Configuration of PHP	07	36
2.	PHP Structure and Syntax	07	
3.	Control Structures	08	
4.	Functions in PHP	08	
5.	Working With Forms	09	
6.	Database Connectivity in PHP	09	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit - I: Introduction and Configuration of PHP Hours: 07

Introduction of open source software, pros and cons of open source software, open source vs. close source software, Introduction to Webpage and Website, Static and Dynamic Website, Client & Server Side Scripting.

Introduction to PHP, Installation of Apache, Mysql and PHP.

Unit - II: PHP Structure and Syntax   Hours: 07

How PHP code is parsed, Embedding PHP and HTML, Executing PHP and viewing in Browser, Data types, Operators, PHP variables: static and global variables, Comments in PHP.

Unit - III: Control Structures   Hours: 08

Condition statements: If...Else, Switch, Ternary operator.

Loops: While, Break Statement, Continue, Do...While, For, For each.

Exit, Die, Return.

Working with Array in PHP.

Unit - IV: Functions in PHP   Hours: 08

Built-in functions: String Functions, Math Functions, Date Functions, Array Functions

User Defined Functions

Unit - V: Working with Forms   Hours: 09

FORM element, INPUT elements, Validating the user input, Passing variables between pages: Passing variables through GET, Passing variables through POST, Passing variables through REQUEST.

Sessions and cookies: Concept of Session, Starting session, Modifying session variables, Un registering and deleting session variable, Concept of Cookies.

Unit - VI: Database Connectivity in PHP   Hours: 09

Introduction to Database connectivity in PHP, MySQL structure, Connecting MySQL with PHP, Performing query operations with MySQL

Core Books:

1. SteverHolzner: The Complete Reference PHP, 1st Edition, Tata McGraw Hill.
2. Leon Atkinson: Core PHP Programming, 3rd Edition, Pearson Publishers.
3. Matt Doyle: Beginning PHP 5.3, Wrox Publication, 2010.

Reference Books:

1. Peter Moulding: PHP Black Book 2001.
2. Jason Garner, Morgan Owens, Elizabeth Naramore,MattWarden,JeremyStolz: Professional LAMP: Linux, Apache, MySQL and PHP Web Development(Paperback), 1st Edition, Wrox Publication, 2005.

Web References:

1. http://www.w3schools.com/php[For PHP Tutorial]
2. http://www.php.net/[For PHP Function Content]
3. http://www.tutorialspoint.com/[For PHP Tutorial]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Gain the skills and knowledge to install and use an integrated PHP/MySQL environment.

C02 : Able to build dynamic web applications with MySQL database connections.

C03 : Learn the request/response cycle, including GET/POST.

C04 : Write PHP code to produce outcomes and solve problems using various available functions and control structures.

C05 : Learn the different state management techniques.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction and Configuration of PHP	✓				
2	PHP Structure and Syntax				✓	
3	Control Structures				✓	
4	Functions in PHP				✓	
5	Working with Forms			✓	✓	✓
6	Database Connectivity in PHP	✓	✓	✓		

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	2	1	2	3	-	-	1
CO2	3	3	1	2	3	-	-	1
CO3	3	3	1	2	3	-	-	1
CO4	2	3	1	2	3	-	-	1
CO5	2	2	1	2	3	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA510 : Multi Paradigm Scripting

(200 Marks)

Contact Hours: 07

Pre-requisite: CA402: Introduction to Programming in C, CA407: Advanced Programming in C, CA503: Object Oriented Programming using C++.

Methodology & Pedagogy: Theory sessions are required to address computational power of python through its ability to deploy programs using functional, object oriented and web based aspect. Practical sessions demonstrate the implementation of the concepts which are taught during the theory sessions. Case study will help the students to come out with one working module in any of the paradigm through python.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	**Practical**
1	Introduction to Python in context of Multi Paradigm Script	10	36
2	Getting Started with Python	08	
3	Python Basics	08	
4	Python Objects, Iterative Programming & File Input /Output	08	
5	Python functional Programming & Mapping & Set Types	07	
6	Exception Handling & Introduction to Client Server Programming	07	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit – I: Introduction to Python in context of Multi Paradigm Script Hours: 10
Overview of programming Paradigms-Imperative, Functional, logic and object oriented Introduction to Python Programming Language, Downloading and Installing, Running Python, Python Documentation. Features of Python and its origin.

Unit – II: Getting Started with Python Hours: 08

Input through standard input device, Generating Output on console, Operators in python, how to use Numbers in python, String and its variations in python, Comment statement in python.

Unit – III: Python Basics Hours: 08

Python various Statement & its Syntax, Python Identifiers, Basic Programming Style of Python, Memory Management scheme adopted in Python, Conditional & Branching Statement in python.

Unit – IV: Python Objects, Iterative Programming & File Input/Output: Hours: 08

Python objects and its variations: List, Tuple & Set

Different looping constructs available in Python: While, for, continue, break and pass statement.File open, close and other file manipulation functions.

Unit – V: Python functional Programming & Mapping & Set Types: Hours: 07

What are functions, calling functions, creating functions, passing functions, formal arguments, variable length arguments, default arguments, returning values from the functions, returning multiple values from the functions, functional programming, variable scope and recursion.

Special Python object like Dictionary, nesting of one object into another/same object, and other built in functions for list, tuple, set and dictionary.

Unit – VI: Exception Handling & Introduction to Client Server Programming: Hours: 07

Exceptions in Python, Detecting & handling exception, Raising exception, Standard exception & Creating exception

Introduction to web programming, Creating Simple Web Client, Advanced Web Clients, CGI, Building CGI Applications

Core Books:

1. Wesley J. Chun: Core Python Programming, 2nd edition, Pentice Hall,2006.
2. Megnus Lie Hetland: Beginning Python from novice to professional, 2nd edition, Apress,2009.

Reference Books:

1. Richard L. Halterman: Learning to Program with Python, Southern Adventist University 2011.
2. Mark Lutz: Programming Python, 4th Edition,O'reilly, 2011. 3.Steve Holden: Python Web Programming, 1st edition,2002.

Web References:

1. https://www.python.org/about/gettingstarted/ *[For beginners of Python]*
2. http://ocw.mit.edu/courses/electrical-engineering-and-computerscience/6-189-agentleintroduction-to-programming-using-pythonjanuary-iap-2008/ [Python Materials]

3. http://people.cs.aau.dk/~normark/prog303/html/notes/paradigms_them esparadigmoverview-section.html [Overview of Programming Paradigms] 4. https://docs.python.org *[Python Documentation]*

Course Outcomes: Upon successful completion of the course, the students will

C01 : Students will able to understand scripting and the contributions of scripting languages.

C02 : Clear Understanding of the built-in objects of Python and object-oriented concepts

C03 : Understand and summarize different File handling operations

C04 : Be exposed to Web based MVC Application.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes			
		C01	**C02**	**C03**	**C04**
1	Introduction to Python in context of Multi Paradigm Script	✓			
2	Getting Started with Python		✓		
3	Python Basics		✓		
4	Python Objects, Iterative Programming & File Input /Output			✓	
5	Python functional Programming & Mapping & Set Types		✓		
6	Exception Handling & Introduction to Client Server Programming				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	-	1	2	1	-	-	-
CO2	3	3	2	2	1	-	-	2
CO3	2	-	-	1	-	-	-	1
CO4	3	2	2	2	2	2	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA511: Foundation of Data Science

(200 Marks)

Contact Hours: 07

Pre-requisite: CA408 Database Fundamentals, CA409 Computer Oriented Numerical Methods

Methodology & Pedagogy: During theory sessions topics related to Data Science will be will be covered with suitable examples. During Practical Sessions, students will be required to develop small applications related to reading, cleaning, analysis and visualization of data using R programming language. The case study will cover the application and future trends of Data Science.

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Data Science Introduction	07	36
2	Basic Statistics for Data Science	09	
3	Reading and Cleaning the Data	08	
4	Data Visualization	09	
5	Exploratory Data Analysis and Regression	08	
6	Data Science Applications and future trends	07	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit –I Data Science Introduction Hours:07
What is Data Science?, Skills for Data Science, Data and its related problems Introduction to image data,Data Collection,Data Cleaning, Data Analysis, Data Visualization, Data Science Tools

Unit – II Basic Statistics for Data Science Hours:09
Statistic as a Science**,** Data Collection
Descriptive Statistics- Graphical Means of Data Description for a single variable, Principle, Graphical Representation of Data, Applications using Data Science tool R, Measures of Cental Tendancy- The mode, The median, The mean, Applications using R Probability- Definition, Conditional Probability and Independence, Random Variable, Applications using R

Unit-III Reading and Cleaning the Data Hours:08
Reading the Data- Text File, CSV file,image data
Cleaning Up the Data- Consistent Data,Detection and Localization of Errors, Missing Values,Special Values, Outliers, Obvious Inconsistencies, Error Localization, Libraries to handle real time data

Unit-IV Data Visualization Hours:09

Introduction**,** Basic Visualization**-** Scatter Plots**,** Bar Plot, Pie Chart, Common Ploting Task
Layered Visualization using ggplot2- Creating plots using qplot() Interactive Visualizations Using Shiny

Unit-V Exploratory Data Analysis and Regression Hours:08

Data Set Size, Summerizing the Data, Ordering Data, Group and Spilt Data, Variable Correlation,Box Plot, Histogram
Regression Models, Simple Regression Model

Unit – VI Data Science Applications and future trends Hours:07

Applications- Internet Search, Digital Advertisement, Recommender Systems, Image and Speech Recognition, Gaming
Future Trends- Machine Learning to rule the industry, IoT to Conquer BI, Big Data Technology

Core books:

1. K.G.Srinivasa,G.M.Siddesh,ChetanShetty,SowmyaB.J:Statistical Programming in R, First Edition, Oxford Press, 2017.
2. Lillian Pearson : Data Science for Dummies, John Willey & Sons,2015.

Reference books:

1. Rachel Schutt, Carthy O'Neil: Doing Data Science, First Edition,O'Reilly, 2014.
2. MurtazaHaider: Getting Started with Data Science Making Sense of Data with Analytics,Pearson

India,2016

Web References:

1. https://www.analyticsvidhya.com/blog/2015/09/applications-data-science/
2. [Applications of Data Science]
3. https://cran.r-project.org/doc/contrib/de_Jonge+van_der_Loo-Introduction_to_data_cleaning_with_R.pdf [Data Cleaning in R]
4. r4ds.had.co.nz/ [Book of Data Science with R]
5. https://www.r-statistics.com/tag/visualization/[Blog of Data Visualization Using R]
6. https://www.youtube.com/watch?v=dMpdoprDEDI[Vidio on Data Science Tutorials for Beginers]
7. http://101.datascience.community/tag/ebook/[Data Science book with non-technical description]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Understand the fundamentals of Data Science.
C02 : Understand the role of statistics for the Data Science.

C03 : Understand the basics of Data Science Process.
C04: Learn the role of visualizations and EDA in Data Science.
C05: Understand the real applications and future of Data Science.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Data Science Introduction	✓				
2	Basic Statistics for Data Science		✓			
3	Reading and Cleaning the Data			✓		
4	Data Visualization				✓	
5	Exploratory Data Analysis and Regression				✓	
6	Data Science Applications and future trends					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	-	-	3	-	-	-	-
CO2	3	2	3	2	-	-	-	-
CO3	2	3	3	3	-	-	-	-
CO4	3	1	1	2	2	3	-	3
CO5	-	-	-	1	3	-	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA512: Fundamentals of Embedded Programming
(Marks 200)

Contact Hours: 07

Pre-requisite:

- CA 407: Advanced Programming in C
- CA 403: Information Technology and Digital Electronics
- CA 406: Operating System Concepts

Methodology & Pedagogy: During theory lectures illustrations of certain real world problems, which are to be solved using Embedded programming will be discussed. Embedded system development tools will be introduced for solving such problems and "embedded C" computer language will be introduced. During Practical sessions, students will be required to Develop Embedded programs in order to solve moderate size real world problems.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction to Embedded System	07	36
2	Microcontroller Architecture	07	
3	Embedded C	09	
4	Experiment with Microcontroller	10	
5	RTOS	08	
6	Application of Embedded System	07	

Total Hours (Theory): 48
Total Hours (Lab):36
Total: 84

Detailed Syllabus:

Unit – I: Introduction to Embedded System Hours: 07

Embedded System, Characteristics of an Embedded System, Embedded System Basics (Embedded system hardware and Software), Challenges for Embedded System, Typical Embedded System, Classification of Embedded System

Languages for Programming Embedded System, Fundamental Components of Embedded System, Skills required for Embedded System

Unit – II: Microcontroller Architecture Hours: 07

Basics of Processor, Types of Processor, Microcontroller and Microprocessor, CISC, RISC, Criteria for Choosing Microcontroller.

Architecture and feature of 8051 Microcontroller, I/O Ports, Modes of timer and serial communication, Pin diagram of 8051, Memory organization.

System on Chip, Register, Semiconductor, GPIO, UART.

Unit – III: Embedded C Hours: 09

Embedded C Compilers,Embedded C vs. ANSI C.

Template for Embedded C Program, steps to Building Embedded C Programming.

C preprocessor Directives:<reg51.h>, <reg52.h>, Programing Time Delays: Loop, Data

Type, Variables in Embedded C, Functions, Decision Making Condition, Arrays. sbit, bit, sfr, hex file.

Unit – IV:Experiment with Microcontroller Hours: 10

GPIO Interfacing and Programming: LED, Seven Segment, Buzzer

Delay Calculation using on chip timer and application, LCD Interfacing and Programming, Serial device interfacing using UART protocol, Data Acquisition system using ADC, Digital and Analog sensor interfacing, Actuators interfacing and programming: DC, Relay, Servo. Digital Sensor, LCD, DC Motor, Stepper Motor, Relay, Buzzer.

Unit – V: RTOS Hours: 08

Inter process communication, RTOS, Classification of RTOS, Difference between GPOS and RTOS, Service provided by RTOS, Features of RTOS.

Selection of RTOS,Impacts of RTOS on development of the design.

RTOS Architecture, Multitasking, Process life cycle, Task States, Context Switch, Scheduling, Algorithm, shared memory problem.

Unit – VI: Emerging trends in embedded systems and applications Hours: 07

Patterns insight from the applications of embedded systems in real life: Multicore in embedded.

Embedded operating systems. Embedded digital security and surveillance.

Convergence embedded systems and applications: Healthcare, Automotive, Entertainment, Localization and internationalization. Real Time Applications of Embedded Systems.

Core Books:

1. Muhammad Ali Mazidi, Janice GillispieMazidi, Rolin D. McKinlay: The 8051 Microcontroller and Embedded Systems: Using Assembly and C, Pearson Education India, 2007
2. K. Shibu: Introduction to Embedded Systems, Tata McGraw-Hill Education

3. K.C. Wang: Embedded and Real-Time Operating Systems, Springer Publication

Reference Books:

1. Raj Kamal: Microcontrollers: Architecture, Programming, Interfacing and System Design, Pearson Education India, 2009.
2. Michael Barr: Programming Embedded Systems in C and C++, O'REILLY.
3. Warwick A. Smith:C Programming for Embedded Microcontrollers, Elektor International Media BV, 2008

Web References:

1. https://www.elprocus.com/basics-and-structure-of-embedded-c-program-withexamplesforbeginners/
2. http://www.radio-electronics.com/info/processingembedded/embeddedsystems/embeddedprocessing-unit.php
3. http://www.eetimes.com/author.asp?section_id=36&doc_id=1287560[Unit-VI]
4. https://www.scribd.com/document/344825325/B122LnotesBV-pdf[Unit-IV]
5. https://www.tutorialspoint.com/embedded_systems/embedded_systems_tutorial.pdf
6. https://www.embeddedrelated.com/showarticle/453.php
7. https://www.elprocus.com/embedded-system-programming-using-keil-c-language/ [LabSession]
8. https://www.renesas.com/en-in/doc/products/tool/apn/res05b0008_r8cap.pdf[RTOS]

Course Outcomes: Upon successful completion of the course, the students will

C01 : Foster ability to understand the internal architecture and interfacing of different peripheral devices with Microcontrollers.
C02 : Ability to write the programs for microcontroller.
C03 : Foster ability to understand the role of embedded systems in industry.
C04 : Foster ability to understand the design concept of embedded systems.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	**C02**	**C03**	**C04**
1	Introduction to Embedded System	✓			
2	Microcontroller Architecture	✓			
3	Embedded C		✓		
4	Experiment with Microcontroller		✓	✓	
5	RTOS			✓	✓
6	Application of Embedded System			✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	-	2	3	-	-	-	-
CO2	3	2	3	2	-	-	-	-
CO3	2	2	1	3	-	-	-	-
CO4	3	1	1	2	2	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA513: Management Information System

(100 Marks)

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During the Theory sessions, illustrations of working real time information system will be demonstrated. Industrial visits will be arranged in order to demonstrate actual functioning of such systems. Students will be required to carry out a case study of such system.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours
		Theory
1.	Introduction to Information System	05
2.	Transaction Processing Systems	07
3.	Management Information System	06
4.	Decision Support System	06
5.	Knowledge Management & Executive Support Systems	07
6.	Introduction To ERP and Technology	05

Total Hours (Theory): 36
Total: 36

Detailed Syllabus:

Unit - I: Introduction to Information System Hours:05
Data & Information, What is a system, Types of Systems, Types of information, Quality of Information, What is an Information System, Functions of Information System, classification of an Information System, Major Types of Information Systems, Interrelationship of Information Systems.

Unit - II: Transaction Processing Systems Hours:07
Introduction of Transaction Processing System (TPS), Transaction Processing System (TPS) Architecture, Transaction Processing System (TPS) Life cycle, Methods of Transaction Processing System (TPS), Information System Availability Control, Illustrations and live demonstration of TPS such as Banking System, Railway reservation system etc.

Unit - III: Management Information System Hours:06
Introduction to Management: Approaches of Management, Function of Management, System from a Functional Perspective, Reports of Management Information System, A business perspective of MIS, Dimensions of MIS, contemporary Approaches to Information System.

Unit - IV: Decision Support System Hours:06
Business value of Improved Decision making, Types of Decision, Decision making Process, The difference between MIS and DSS, Components of DSS, System for Decision Support Group Decision Support System, Business value of GDSS.

Unit - V: Knowledge Management & Executive Support Systems Hours:07
Important Dimensions of knowledge, Organizational learning and Knowledge Management, The Knowledge Management value chain, Overview of different types of Knowledge Management Systems, Characteristics of ESS, The Role of ESS in the Firm, Business value of ESS, Monitoring Corporate Performance, Working Examples of ESS.

Unit - VI: Introduction To ERP and Technology Hours:05
An overview of ERP, Basic ERP Concepts, Risk and Benefits of ERP, ERP and Related Technologies, Business Intelligence

Core Books:

1. K. C. Laudon and J. P. Laudon: Management Information Systems, 12th Edition, Pearson Education.
2. Alexis Leon : ERP Demystified, Second Edition, Tata McGraw-Hill.

Reference Books:
1.W.S. Jawadekar: Management Information Systems, 2nd Edition, TataMcGraw-Hill.

Web References:

1. http://www.slideshare.net/NorazilaMat1/laudon-mis12-ppt01-16595885[Function of Information system, A business perspective of MIS, Dimensions of MIS, Contemporary Approaches to Information System]
2. http://www.uh.edu/~mrana/try.htm[Types and Functions of Information system]
3. http://bisom.uncc.edu/courses/info2130/Topics/istypes.htm[Types of Information system]
4. http://kalyan-city.blogspot.com/2011/05/levels-of-management-top-middleand.html[classification of information system]
5. http://my.safaribooksonline.com/book/management/9780470916803/operationalplanningandcontrol-systems/management_levels_comma_functions_comm[classification of information system]
6. http://ambarwati.dosen.narotama.ac.id/files/2011/05/FIS-2011-w2.pdf[Transaction Processing System]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : To understand basic of data & information, types of information system ,function of information system and its quality.

C02 : To understand transaction processing concepts and its working.

C03 : Learning Management Information system, its types and its implementation.

C04 : To gain understanding of Decision Support Systems.

C05 : To learn Knowledge Management, Executive Support Systems, ERP and its technology.

Course Outcomes Mapping :

Unit No.	**Unit Name**	**Course Outcomes**				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to Information System	✓				
2	Transaction Processing Systems		✓			
3	Management Information System	✓	✓	✓		
4	Decision Support System		✓		✓	
5	Knowledge Management & Executive Support Systems		✓			✓
6	Introduction To ERP and Technology		✓			✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	2	-	2	-	3	-	3
CO2	3	2	2	-	2	2	2	-
CO3	-	-	3	2	-	2	3	2
CO4	-	-	2	3	3	2	-	-
CO5	-	-	2	2	3	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CL142.01: ENVIRONMENTAL SCIENCES Credits and Hours:

<table>
<tr><th colspan="4">Teaching Scheme</th><th colspan="6">Examination Scheme</th></tr>
<tr><th colspan="3">Contact Hours</th><th>Credit</th><th colspan="3">Theory</th><th colspan="2">Practical</th><th>Total</th></tr>
<tr><th>Theory</th><th>Pract</th><th>Total</th><th></th><th colspan="2">Internal</th><th>External</th><th>Internal</th><th>External</th><th></th></tr>
<tr><td>-</td><td>2</td><td>2</td><td>2</td><td>-</td><td>-</td><td>-</td><td>30</td><td>70</td><td>100</td></tr>
</table>

A. Course Objectives:

1. To impart basic knowledge about environment and there by developing an attitude of concern towards environment.
2. To inculcate alertness towards environment.
3. To make awareness on delineating on various environmental pollution and their effects on environment.
4. To deliver a comprehensive insight into natural resources, ecosystem mand biodiversity.
5. To develop the curiosity and visionary of student in relation to environment.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	05
2	Environmental Pollution	12
3	Ecology &Ecosystems	10
4	Natural Resources	03

Total Hours(Theory):30
Total Hours(Lab):00
TotalHours:30

Outline of the Course:

1 Introduction 05Hours 24%

1.1 Basic definitions

1.2 Objectives and guiding principles of environmental studies

1.3 Components of environment

1.4 Structures of atmosphere

1.5 Man-Environment relationship

1.6 Impact of technology on the environment

2 Environmental Pollution 12Hours 33%

2.1 Environmental degradation

2.2 Pollution, sources of pollution, types of environmental pollution

2.3 Air pollution: Definition, sources of air pollution, pollutants, classifications of air pollutants(common likeSO_X&NO_X), sources &effects of common air pollutants

2.4 Water pollution: Definition, sources water pollution, pollutants & classification of water pollutants, effects of water pollution, eutrophication

2.5 Noise pollution: Sources of noise pollution, effects of noise pollution.

2.6 Ill Effects of Fireworks: Severity of toxicity, environmental effects and health hazards.

2.7 Current environmental global issues, global warming &green houses, effects, acid rain, depletion of Ozone layer

3 Ecology &Ecosystems 10Hours 33%

3.1 Ecology: Objectives and classification

3.2 Concept of an ecosystem: Structure& function

3.3 Components of ecosystem: Producers, consumers, decomposers

3.4 Bio-Geo-Chemicalcycles & its environmental significance

3.5 Energy flow in ecosystem

3.6 FoodChains: Types & food webs

3.7 Ecological pyramids

3.8 Major ecosystems

4 Natural Resources 03 Hours 10%

4.1 Natural resources: Renewable resources, nonrenewable resources, destruction Versus conservation

4.2 Energy resources: Conventional energy sources & its problems, nonconventional energy sources-advantages &its limitations, problems due to over exploitation of energy resources

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the course will be discussed.
- Lectures will be conducted with the aid of multi-media projector, blackboard, OHP etc.
- Attendance is compulsory in lectures which carries 10 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignment/Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.

E. Course Outcomes:

On the successful completion of the course the students will be able

CO1 :To perceive the elementary knowledge about natural environment and its relation with science.

CO2 :To identify and analyze human impacts on the environment.

CO3:To understand the facts and concepts of natural and energy resources thereby applying them to lessen the environmental degradation.

CO4 :To communicate on recent environmental problems thereby creating awareness among society.

F. Recommended Study Material:

Text Books:

1. Varandani, N.S., Basics of Environmental Studies
2. Sharma, J.P., Basics of Environmental Studies

ReferenceBooks:

1. Shah Shefali & Goyal Rupali, Basics of Environmental Studies
2. Agrawal,K.C., Environmental Pollution: Causes, Effects& Control
3. Dameja, S.K., Environmental Engineering & Management
4. Rajagopalan, R., Environmental Studies, Oxford University Press
5. Wright Richard T. & Nebel Bernard J., Environmental Science
6. Botkin Daniel B. & Edward A. Keller, Environmental Science
7. S.G.,Shah, G.N. Shah, Basics of Environmental Studies, Superior Publications, Vadodara

Web Materials:

- http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-Delhi/Environmental%20Air%20Pollution/index.htm
- http://nptel.iitm.ac.in/video.php?subjectId=105104099
- http://apollo.lsc.vsc.edu/classes/met130/notes/chapter1/vert_temp_all.html
- http://www.epa.gov
- http://www.globalwarming.org.in
- http://nopr.niscair.res.in
- http://www.indiaenvironmentportal.org.in

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	Po8
CO1	-	2	-	2	-	-	3	-
CO2	-	2	-	2	-	-	3	-
CO3	-	2	-	2	-	-	3	-
CO4	-	2	-	2	-	-	3	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS133 C
Creativity, Problem Solving and Innovation (Sem-IV)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
IV	HS133 A/B/C/D/E/F/G/H	Creativity, Problem Solving and Innovation	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Introduction to Creativity, Problem Solving and Innovation** • *Definitions of Creativity and Innovation* • *Need for Problem Solving and Innovation* • *Scope of Creativity in various Domains* • *Types and Styles of Thinking* • Strategies to develop Creativity, Problem Solving and Innovation skills	**06**
2	**Questioning, Learning and Visualization** • *Strategy and Methods of Questioning* • *Asking the Right Questions* • *Strategy of Learning and its Importance* • *Sources and Methods of Learning* • *Purpose and Value of Creativity Education in real life* • *Visualization strategies - Making thoughts Visible* • *Mind Mapping and Visualizing Thinking*	**06**
3	**Creative Thinking and Problem Solving** • Creative Thinking and its need • Strategy of Thinking Fluency • Generating all Possibilities • SCAMPER Technique • Divergent Vs Convergent Thinking • Lateral Vs Vertical Thinking • Fusion of Ideas for Problem Solving • Applying strategies for Problem Solving	**06**

4	**Logic, Language and Reasoning** • *Basic Concepts of Logic* • *Statement Vs Sentence* • *Premises Vs Conclusion* • *Concept of an Argument* • *Functions of Language: Informative, Expressive and Directive* • *Inductive Vs Deductive Reasoning* • *Critical Thinking & Creativity* • *Moral Reasoning*	**06**
5	**Contemporary Issues and Practices in Creativity and Problem Solving** • *Cognitive Research Trust Thinking for Creatively Solving Problems* • *Case Study on Contemporary Issues and Practices in Creativity and Problem Solving*	**06**
	Total	**30**

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Puzzles, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc.

IV. Evaluation:

There will be no end semester university examinations. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation. The concerned teacher shall evaluate students distribute the marks (out of 30 as Internal and out of 70 as External) and submit them.

Evaluation Scheme

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Attendance	100 %	--	20
2	Individual Activity Participation	As stipulated by the Resource Person(s) in the Training		20
3	Group Activity Participation			20
4	Presentation			30
5	Feedback on Improvement			10
			Total	**100**

Course Outcome (COs):

After completion of the course, the student would :

CO1 Demonstrate creativity in their day to day activities and academic output

CO2 Solve personal, social and professional problems with a positive and an objective mindset

CO3 Think creatively and work towards problem solving in a strategic way

CO4 Initiate new and innovative practices in their chosen field of profession

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	3	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Books / Reference Books / Reading

Text Books

V. R Keith Sawyer, Zig Zag, The Surprising Path to Greater Creativity, Jossy-Bass Publication 2013

VI. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

Reference Books

3. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
4. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
5. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
6. Tony Buzan, How to Mind Map, Thorsons Publication 2002
7. Scott Berkum, The Myths of Innovation, Expended and revised edition, Berkun Publication 2010
8. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
9. Ira Flatow, The all Laughed, Harper Publication 1992
10. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

Additional Readings

11. Keith Sawer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
12. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
13. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
14. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
15. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
16. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
17. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012

18. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
19. Soctt G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
20. Donald J Treffinger, scott G Isaksen, K Brain stead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
21. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
22. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
23. Howard Gardner, Creating minds, Basic Books Publication 1993
24. Mihaly Csikzentmihalyi, Creativity–Flow and Psychology of Discovery and Invention, Harper Publication 1996
25. Martin Gerdner, W. H., Ahal Insight, Freeman Publication 1978
26. Paul Sloane, Test Your Lateral Thinking IQ, Sterling Publication1994
27. Paul Sloane & Des Machale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

Articles / Videos / Other Suggested Materials

- Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

Teaching and Examination Scheme (B.Sc.(IT) Programme) Effective from Year 2019-20

Semester-V

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA601	ObjectOriented Programming Through JAVA	4	3	7	7	10	20	70	15	15	70	200
CA602	Application development using .NET Framework	4	3	7	7	10	20	70	15	15	70	200
CA603-607	Elective – II	4	3	7	7	10	20	70	15	15	70	200
CA608	Object Oriented System Analysis and Design	-	3	3	3	-	-	-	30		70	100
HS124.01 C	Professional Communication	-	2	2	2	-	-	-	30		70	100
		12	14	26	26	300			500			800

Elective-II

1. CA603 Fundamentals of Multi-Media
2. CA604 Open Source Framework - I
3. CA605 Scripting Framework - I
4. CA606 Advanced Data Science
5. CA607 Embedded Programming Using Arduino

CA601 Object Oriented Programming Through JAVA

(200 Marks)

Contact Hours: 07

Pre- requisite: Basic knowledge of Introduction to Programming in C and OOPs concepts.

Methodology & Pedagogy: Uponsuccessful completion of the syllabus, students will get basics of object oriented programming. Concretely, students shall be able to create appropriate classes using the Java Programming Language to solve problems using Object Oriented Approach. They shall be able to write console based and multithreaded applications using the Java Programming Language.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Basics of Java	07	36
2	Building blocks of the Language	06	
3	Object Oriented Programming Concepts	12	
4	Exception Handling and String Handling	07	
5	Packages and Interfaces	08	
6	Multithreaded Programming	08	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit – I: Basics of JavaHours : 07

Basics of Java, Background/History of Java, Advantages of Java, Java Virtual Machine & Byte Code, Java Environment Setup, Java Program Structure, differentiate between POP and OOP, Basics of OOP: Abstraction, Inheritance, Encapsulation, Classes, subclasses and super classes, Polymorphism and Overloading, write simple programs using java.

Unit – II: Building blocks of the LanguageHours : 06

Data types: constant and variables, Garbage Collection, Operators, Decision & Control Statements

Unit – III: Object Oriented Programming Concepts Hours :12

Defining classes, fields and methods, creating objects, accessing rules, this keyword, static keyword, method overloading, final keyword,Constructors: Default constructors, Parameterized constructors, Copy constructors, Passing object as a parameter, constructor overloading, Basics of Inheritance, Types of inheritance: single, multiple, multilevel, hierarchical and hybrid inheritance, concepts of method overriding, extending class, super class, subclass, dynamic method dispatch & Object class.

Unit – IV: Exception Handling, String HandlingHours :07

String class and operations, StringBuffer class and operations and StringBuilder class and operations. Exception Handling: types of exceptions, Throwable class, keywords – try, catch, throw, throws and finally, nested try statement, java built in exceptions, user defined exceptions.

Unit – V: Packages and InterfacesHours : 08

Creating package, importing package, access rules for packages, class hiding rules in a package. Exploring java.lang package: Wrapper classes and Simple Type Wrappers, Void, Runtime Class, System Class, Using clone() and Colable() interface, Math Class. Exploring java.util package: Random, StringTokenizer, Calendar, Date and Enumeration classes. Defining interface, inheritance on interfaces, implementing interface, multiple inheritance using interface. Abstract class and final class.

Unit – VI: Multithreaded Programming Hours :08

The java Thread Model, priorities, Thread class, Runnable interface, creation of Thread, creating multiple threads, synchronization and deadlock.

Core Books:

1. Herbert Schildt: The Complete Reference Java J2SE, 5th Edition, TMH Publishing Company Ltd.
2. Sachin Malhotra, SaurabhChodhary: Programming in Java, 2nd Edition, OXFORD press.

Reference Books:

1. Pravin Jain: The class of JAVA, 3rd Impression, Pearson.
2. Ivor Horton: Beginning Java 2" JDK, 5th Edition, Wiley Computer Publishing 2007.
3. Ken Arnold, James Gosling and David Holmes: The Java Programming Language, 4th Edition, Addison Wesley.

Web References:

1. http://www.tutorialspoint.com/java/ [For all JAVA Concepts]
2. http://www.roseindia.net/java/ [For JAVA Tutorials with Example]
3. https://www.javatpoint.com/java-tutorial/[For all JAVA Concepts]

Course Outcomes: Upon successful completion of the course, the students will:

C01: Basic building blocks of language

C02 : To introduce the object oriented programming concepts.

C03 : To understand object oriented programming concepts, and apply them in solving Problems.

C04 : To introduce the principles of inheritance and polymorphism; and demonstrate how they relate to the design of abstract classes

C05 : To introduce the implementation of packages and interfaces

C06 : To introduce the concepts of exception handling and multithreading.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Basics of Java	✓					
2	Building blocks of the Language	✓	✓				
3	Object Oriented Programming Concepts		✓	✓	✓		
4	Exception Handling, String Handling			✓			✓
5	Packages and Interfaces					✓	
6	Multithreaded Programming			✓		✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	2	2	1	-	2
CO2	3	3	3	2	2	1	-	2
CO3	3	3	3	3	2	2	2	3
CO4	3	3	2	3	2	1	1	3
CO5	2	2	3	3	2	1	1	1
CO6	2	3	3	3	2	1	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA602 Application Development using .NET Framework

(Marks 200)

Contact Hours: 07

Pre- requisite: None.

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for advanced features of .Net framework and to develop windows based application using C# language will be given. During Practical sessions, students will be required to develop Web Applications using concepts discussed during class.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction to .NET framework	08	36
2	Basics of C# language	08	
3	File Management	06	
4	Building desktop application	10	
5	Database connectivity through ADO.NET	08	
6	Building WPF application	08	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit – I: Introduction to .NET framework Hours: 08

Introducing .NET Framework, .NET Framework Component, .NET Framework Version Compatibility, Core of .NET Framework: Application Services, Base Class Library and CLR, Types of Applications which can be developed using MS.NET, MS.NET Base Class Library, MS.NET Namespaces, The Common Language Runtime (CLR), Managed Code, MS.NET Memory Management / Garbage Collection, Common Type System (CTS), Common Language Specification (CLS) , Types of JIT Compilers

Unit – II: Basics of C# language Hours: 08

C# Fundamentals: Basic classes, declarations, conditionals, loops, arrays, strings, enumerations, structures, and Reference Type and Value Type, Datatypes &

Variables Declaration, Implicit and Explicit Casting, Checked and Unchecked Blocks – Overflow Checks,
Casting between other datatypes, Boxing and Unboxing, Enum and Constant, Operators, Control Statements, Working with Arrays, C# Application Basics: Command line and VS.NET compilation.
Developing Console Application: Introduction to Project and Solution in Studio, Entry point method – Main, Compiling and Building Projects, Using Command Line Arguments, Importance of Exit code of an application, Different valid forms of Main, Compiling a C# program using command line utility CSC.EXE

Unit – III: File Management Hours: 08

Introduction to file handling classes, Creating, Reading and Writing files, Manipulation operation on files.

Unit – IV: Building desktop application Hours: 10

Building Windows Forms Applications, Setting Form Properties, Understanding the Life cycle of a Form, Using the Windows Forms Designer, MessageBox Class and .config File, Working with
Windows Forms Controls: TextBox, Button, Selection, List, Container, Image, ErrorProvider, TooltipProvider, etc. Handling Events: Understanding the Event-Driven Programming Model, Writing Event Handlers, Sharing Event Handlers

Unit – V: Database Connectivity through ADO.NET Hours: 08

Evolution of ADO.NET, Understanding the ADO.NET Object Model, Connected vs. Disconnected Access, Use of Connection, Command, DataReader, DataAdapter and Dataset class. Understanding ADO.NET Data Binding: Binding to Simple and Complex Controls, Manually Binding Controls, Use of BindingSource Control, BindingNavigator Control and datagridview control.

Unit – VI: Building WPF Application Hours: 06

Overview of WPF, Features of WPF, Windows Forms vs WPF , Introduction to XAML
WPF Layout: Grid Panel, Stack Panel, Dock Panel, Wrap Panel and Canvas Panel
WPF Controls: TextBlock, PasswordBox, ListBox, Slider, Popup, Menus and Expander, 2D Animation in WPF.

Core Books:

1. Christian Nagel, Bill Evjen, Jay Glynm, Karli Watson, Morgan Skinner: Professinal C# 2012 and .NET 4.5, Wiley – India, WROX
2. .NET 4 for Enterprise Architects and Developers
3. AlbahaniJosheph: C# 4.0 Pocket Reference, Shroff publication,2010

Reference Books:

1.Drayton David: C# Language Pocket Reference, Shroff publication,2005

Web References:

1. http://csharp.net-informations.com/ [to learn C# and .NET fundamentals]
2. http://www.wpftutorial.net/[to learn WPF]
3. http://www.c-sharpcorner.com/UploadFile/puranindia/file-handling-in-C-Sharp/[File Management]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Understanding .NET Framework core features and Components.
C02 : To develop console based applications and understanding the relation between core language dependency.
C03 : To learn how to manage files with C# language in console based applications
C04 : To develop windows based application using core feature of .NET framework and C# language.
C05 : Access and manipulate data in a Microsoft SQL Server database by using Microsoft ADO.NET.
C06 Use modern approach to build windows based application using WPF.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes					
		C01	**C02**	**C03**	**C04**	**C05**	**C06**
1	Introduction to .NET framework	✓					
2	Basics of C# language	✓					
3	File Management		✓				
4	Building desktop application			✓			
5	Database connectivity through ADO.NET				✓		
6	Building WPF application					✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	-	-	1	3	-	1	-
CO2	-		3	3	3	-	1	-
CO3	-	2	3	3	2	1	-	-
CO4	3	-	-	3	-	1	2	3
CO5	3	3	3	3	-	1	1	3
CO6	2	2	2	2	1	1	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA603: Fundamentals of Multimedia

(200 Marks)

Contact Hours: 07

Pre-requisite: CA501: Programming the Internet.

Methodology & Pedagogy: During theory sessions topics related to common technologies and techniques used in the multimedia will be covered with suitable examples. During Practical sessions, students will be required to develop and produce animations using appropriate tool.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	**Practical**
1	Introduction to Multimedia	06	36
2	Using Text in Multimedia	07	
3	Working With Images	07	
4	The Internet and Multimedia	09	
5	Learning Web Animations - I	09	
6	Learning Web Animations - II	10	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit – I Introduction to Multimedia Hours:06

What is Multimedia? Defining the scope of multimedia. Usage of Multimedia, Stages of a Multimedia Project, the Multimedia Team.

Unit – II Using Text in Multimedia Hours:07

Introduction, About Fonts and Faces, About Fonts and Faces, Computers and Text, Font Editing and Design Tools, Hypermedia and Hypertext

Unit – III Working With Images Hours:07

Introduction, About Fonts and Faces, About Fonts and Faces, Computers and Text, Font Editing and Design Tools, Hypermediaand Hypertext

Unit – IV The Internet and Multimedia Hours:09

Internet History, Internetworking: Internet Addresses, Connections, The Bandwidth Bottleneck, Internet Services, MIME-Types, The World Wide Web and HTML, Multimedia on the Web: Tools for the World Wide Web, Web Servers, Web Browsers, Search Engines, Web Page Makers and Site Builders, Plug-ins and Delivery Vehicles, Beyond HTML

Unit – V Learning Web Animations-I Hours:09

CSS3 Fundamentals,CSS3 Transforms and Transitions,CSS3 Transitions for Images,CSS3 Transitions for UI Elements

Unit – VI Learning Web Animations-II Hours:10

CSS3 Keyframe Animations,CSS3 Keyframe Animations for Web Content,Integrating CSS3 Animations with SVG and Filters,CSS3 3D Transforms, Transitions, and Animations,Moving Between Landscapes,Tools, Technologies, and the Future of CSS Animation

Core Books:

1. Multimedia: Making it work by Tay Vaughan - TMH, Eighth edition. 200
2. Principles of Multimedia by Ranjan Parekh. Tata McGraw-Hill
3. Pro CSS3 Animation by Dudley Storey.Apress Publisher, 2012

Reference Books:

1. Multimedia systems by John F. Koegal Buford-Pearson Education.
2. Multimedia Systems Design by Prabhat K. Andleigh and Kiran Thakrar-PHI publication
3. Learning CSS3 Animations and Transitions: A Hands-on Guide to Animating in CSS3 with Transforms, Transitions, Keyframes, and JavaScript by Addison-Wesley Professional, 2012

Web References:

1. https://www.w3schools.com/css/css3_animations.asp[CSS3 Animations]
2. https://www.tutorialspoint.com/css/css3_animation.htm[CSS3 Animations]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Be able to identify the basic concepts of multimedia and the web.

C02: Be able to use various multimedia tools.

C03 : Be able to design and create interactive multimedia based applications.

C04 : Able to create animated banner ads using CSS3

C05 : Apply end-user design skills of different application software in creative industry of digital media.

Course Outcomes Mapping:

Unit No.	Content	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to Multimedia	✓				
2	Using Text in Multimedia		✓			
3	Working With Images		✓			
4	The Internet and Multimedia	✓		✓		✓
5	Learning Web Animations - I				✓	✓
6	Learning Web Animations - II				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	-	1	2	2	2	-	2
CO2	3	2	1	2	2	2	-	2
CO3	3	2	1	2	2	2	-	2
CO4	3	-	1	2	2	2	-	2
CO5	3	-	1	2	2	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA604 Open Source Framework-I
(200 Marks)

Contact Hours: 07

Pre-requisite: CA509 Open Source Technology

Methodology & Pedagogy: During the theory lectures, concepts for MVC Framework will be discussed. Also application creation using MVC will be taught. The web applications used in the real world will be discussed with necessary examples. During the laboratory hours' students will implement the concepts that are discussed during lecture.

Outline of the Course:

Unit. No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to code igniter	07	36
2	Introduction to Model View Controller	08	
3	Form Handling	08	
4	Database Connectivity	10	
5	File Uploading & Pagination Class	08	
6	Session Management & Security	07	

Total Hours (Theory): 48

Total Hours (Lab): 36 Total: 84

Detailed Syllabus:

Unit-I: Introduction to Code Igniter Hours: 07

Introduction of code igniter, installation and architecture, Advantages and disadvantages of Code Igniter, Explore the file structure, Code Igniter syntax rules, Types of file or classes on a CI site

Unit-II: Introduction to Model View Controller Hours: 08

Controller: The Business logic, defining a default controller, remapping function calls
Models: Data abstraction layer, Loading a model, connection to your database automatically View: Template files, loading view and multiple views, adding dynamic data, Creating Loops, Returning views as data.

Unit-III: Form Handling Hours: 08

CodeIgniter URLs, URI Routing, Input Class, Form Validation Class, File Uploading Class, Security Class, Page Redirection, Helper functions: Cookie Helper, Date Helper, Download Helper, URL Helper

Unit-IV: Database ConnectivityHours: 10

Database Configuration, connecting to a Database, Running Queries, Generating Query Results, Query Helper Functions, Field Data, Selecting Data, Inserting Data, Updating Data, Deleting Data, Method Chaining

Unit-V: File Uploading & Pagination Class Hours: 08

Create the upload views, create the upload controller: Initial Controller, file type and file size, uploading the file, Pagination: Customizing the pagination, first, last, next, previous, current page and Digit Link

Unit-VI: Session Management & Security Hours: 07

Intilization & Autoloading Session Class, Sending Emails using CI, SMTP and Google, URI Security

Core Books:

1.CodeIgniter 1.7 Professional Development by Adam Griffiths [PACKT] Publishing

Reference Books:

1.Codeigniter for Rapid Php Application Development by David Upton [PACKT] Publishing
2.CodeIgniter 2 Cookbook by Rob Foster [PACKT] Publishing

Web References:

1. https://www.tutorialspoint.com//codeigniter/index.htm
2. https://www.codeigniter.com/user_guide/

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Be able to configure and install the code igniter.

C02 : Able to create MVC based application using the code igniter.

C03 : Able to develop dynamic web application and database communication in the web application development.

C04 : Able to divide long result set data into number of pages through pagination technique. Also able to upload a different types of files in the web page.

C05 : Learn various state management techniques.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to code igniter	✓				
2	Introduction to Model View Controller	✓	✓			
3	Form Handling		✓	✓	✓	
4	Database Connectivity			✓	✓	
5	File Uploading & Pagination Class			✓	✓	
6	Session Management & Security			✓	✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	1	1	2	2	2	-	2
CO2	3	2	1	2	2	2	-	2
CO3	3	2	1	2	2	2	-	2
CO4	3	-	1	2	2	2	-	2
CO5	3	-	1	2	2	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA605: Scripting Framework – I
(200 Marks)

Contact Hours: 07

Pre-requisite: CA510 Multi Paradigm Scripting

Methodology & Pedagogy: Theory sessions are required to address computational power of python through its ability to deploy programs using functional, object oriented and web based and network programming based aspect. Practical sessions demonstrate the implementation of the concepts which are taught during the theory sessions. Case study will help the students to come out with one working application using scripting framework through python.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Object Oriented Programming using Python	08	36
2	Regular Expression	07	
3	Database	08	
4	Network Programming	08	
5	Django Framework	10	
6	Database Handling using Django Framework	07	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit – I: Object Oriented Programming using Python Hours: 08

Classes, Classes attributes, Instances, Instance attributes, Binding and Method Invocation, Static and Class methods, Inheritance, Built-in functions for classes, instances and other objects.

Unit – II: Regular Expression and Iterator Objects Hours: 07

Regular Expression - Defining, Compiling, Using Regular Expression, using match objects to extract a value, extracting multiple items, replacing multiple items.
Iterator Objects- Generator function, class containing generator method, Iterator class, iterator class that uses yield, generator expression.

Unit – III: Database Hours: 08

Introduction, Python Database API- Global Variable, Exceptions, Connections and Cursors, Types.

Unit – IV: Network Programming Hours: 08

Introduction, Socket: Communication endpoints, Network programming in Python, SocketServer module

Unit – V: Django Framework Hours: 10

Introduction, Installing Django, Django and Python, View and template, Basics of dynamic web pages.

Unit – VI: Database Handling using Django Framework Hours: 07

Configuring the database, Basic data access, Inserting and Updating the data, Selecting Objects, Deleting Objects, Make changes to database schema.

Core Books:

1. Wesley J. Chun: Core Python Programming, 2nd edition, Pentice Hall,2006.
2. Megnus Lie Hetland: Beginning Python from novice to professional, 2nd edition, Apress,2009.
3. Adrian Holovaty, Jacob K. Moss: Django- The Definitive Guide to Django: Web Development Done Right, 2nd edition, Apress, 2009.

Reference Books:

1. Richard L. Halterman: Learning to Program with Python, Southern Adventist University 2011.
2. Mark Lutz: Programming Python, 4th Edition,O'reilly, 2011. 3.Steve Holden: Python Web Programming, 1st edition,2002.

Web References:

1. https://www.python-course.eu/sql_python.php*[For database connectivity using Python]*
2. https://wiki.python.org/moin/BeginnersGuide?action=AttachFile&do=view&target=Pyt h onDoc %26Start16DEC2010.pdf*[Python Documentation]*
3. http://www.davekuhlman.org/python_book_01.pdf*[For Regular Expression and Iterator Object]*

4. http://gsl.mit.edu/media/programs/south-africasummer2015/materials/djangobook.pdf*[Django Framework]*

Course Outcomes : Upon successful completion of the course, the students will :

C01 : Understand the concepts of Scripting Framework using Python.

C02 : Be able to design different mode of application like database , network and Web application using Scripting framework.

C03 : Understand how to use Django Framework for real application.

C04 : Be able to handling database using Scripting Framework.

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**			
		C01	**C02**	**C03**	**C04**
1	Object Oriented Programming using Python	✓			
2	Regular Expression	✓			
3	Database	✓	✓		✓
4	Network Programming	✓	✓		
5	Django Framework			✓	
6	Database Handling using Django Framework			✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	1	1	2	2	2	-	2
CO2	3	2	1	2	2	2	-	2
CO3	3	2	1	2	2	2	-	2
CO4	3	1	1	2	2	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA606: Advanced Data Science
(200 Marks)

Contact Hours: 07

Pre-requisite: CA511 Foundation of Data Science

Methodology & Pedagogy: During theory session topics related to advanced data science, such as Big Data, Statistical Inference, Machine Learning and its Applications and Distributed Data Storage and Processing will be covered with suitable examples. Case studies related to application of Data Science will be assigned to the students in the group. During Practical Sessions, students will be required to develop the applications of Data Science using appropriate tools such as R. Distributed Data Processing Architecture HADOOP will also cover in the practical session.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Data Science in Big Data World	07	36
2	Data Science Process	09	
3	Statistical Inference	08	
4	Introduction to Machine Learning	09	
5	Usage and Applications of Machine Learning	08	
6	Distributed Data Storage and Processing	07	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit –I Data Science in Big Data World Hours:07

Usage and benefits of data science and Big Data, Facets of data- Structured data, unstructured data, Natural Language, Machine Generated data, Graph-based or network data, Audio, image and video, streaming data.

Unit – II Data Science Process Hours:09

Setting Research goals, Retrieving data, Data Preparation, Data Exploration, Data Modeling and Model Building, Presentation and Automation

Unit-III Statistical Inference Hours:08

Populations and samples, Statistical modeling, probability distributions, fitting a model, Introduction to R

Unit-IV Introduction to Machine Learning Hours:09

What is Machine Learning? The Modeling Process, Types of Machine Learning- Supervised, Unsupervised and Semi supervised

Unit-V Usage and Applications of Machine Learning Hours:08

Recommendation Systems- Ingredients of Algorithms, Dimensionality Reduction, Singular Value Decomposition, Principal Component Analysis

Social Network- Social Network as a Graph, Clustering of Graphs, Neighborhood Properties in Graph

Unit – VI Distributed Data Storage and Processing Hours:07

Introduction of Distributed Data Processing, Promises of DDBs, Complicating Factors, and Problem Areas, Distribution Design Issues, Fragmentation, Distribution,Transparency, Impact of distribution on user queries, and Allocation

Core books:

1. Davy Cielen,Arno D.B. Meysman, Mohmed Ali : Introducing Data Science, Dreamtech Press, 2016.
2. M. Tamer Ozsu, Patrick Valduriez : Principles of Distributed Database Systems, 2nd Edition, Pearson,2006.

Reference books:

1. Cathy O'Neil and Rachel Schutt : Doing Data Science, Straight Talk From The Frontline, O'Reilly,2014.
2. Bart Baesens :World: TheEssential Guideto Data Science and its Applications, Wiley,2014.

Web References:

1. http://bedford-computing.co.uk/learning/wp-content/uploads/2016/09/introducingdatasciencemachine-learning-python.pdf[Core Book of Data Science]
2. http://pacha.hk/analisis-de-datos-unab/lecturas/ingles/LittleInferenceBook.pdf[Statistical Inference for Data Science]
3. http://alex.smola.org/drafts/thebook.pdf[Introduction to Machine Learning]
4. https://www.isical.ac.in/~acmsc/WBDA2015/slides/hg/Oreilly.Hadoop.The.Definitive.Guide.3rd.Edit ion.Jan.2012.pdf[HADDOP Guide]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Understand the Data Science in perspective of Big Data.
C02 : Understand the process of Data Science in details.
C03 : Understand the role of statistics in Data Science research.

C04 : Learn machine learning techniques and how to use in context to real applications.
C05 : Understand the role and challenges of Distributed Data Storage and processing.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Data Science in Big Data World	✓				
2	Data Science Process		✓			
3	Statistical Inferences			✓		
4	Introduction to Machine Learning				✓	
5	Usage and Applications of Machine Learning				✓	
6	Distributed Data Storage and Processing					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	2	3	3	3	2	2
CO2	2	2	1	2	3	3	1	1
CO3	2	2	3	2	3	2	1	1
CO4	1	2	1	2	2	1	2	2
CO5	2	3	2	1	1	1	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA607: Embedded Programming using Arduino

(Marks 200)

Contact Hours: 07

Pre-requisite: CA512: Fundamentals of Embedded Programming

Methodology & Pedagogy: This is a hands-on course on the theory and practice of developing real-time and embedded systems. The course provides an integrated approach to develop low-power systems with hardware, software, sensors and actuators.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	**Practical**
1	Introduction to Arduino and Development Board	07	36
2	Arduino Hardware and Software	07	
3	Arduino Programming	09	
4	GPIO Programming	09	
5	Sensor and Actuators	08	
6	Wireless Communications using Arduino	08	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit – I: Introduction to Arduino and Development Board Hours: 07
What is Arduino, Why Arduino, what is Development Board, Different Arduino Boards, Arduino UNO board, Memory Scope of Arduino in Embedded Systems

Unit – II: Arduino Hardware and Software Hours: 07
The Arduino Environment, Board Type, Serial Port / COM Port, The Environment, Parts of the Sketch, IDE-Integrated Development Environment, Proteus Arduino, Set Up Arduino in computer, MIT App Inventor

Unit – III: Arduino Programming Hours: 09

Digital I/O: pinMode(), digitalWrite(), digitalRead(), Time:millis(), micros(), delay(), delayMicroseconds(), Input Vs Output, Analog I/O:analogRead(), analogReference(), analogWrite(), Serial.Begin(), Serial.print(), randomSeed, and random()

Unit – IV:GPIO Programming Hours: 09

Blinking LED, Resister, Buzzer, Replays, Toggle, Character LCD 16x2 interfacing logic and concept, Introduction of LCD command and data signals, LCD based programming, Serial Communication programming

Unit – V: Sensor and Actuators Hours: 08

Introduction to Sensors & Actuators, Digital On/Off Sensors, Digital Sensor, Pulse Sensor

Unit – VI: Wireless Communications using Arduino Hours: 08

Introduction to Wireless Devices, Interfacing GSM & GPS Module with Arduino, Interfacing Bluetooth Module with Arduino

Core Books:

1. Mario Bohmer: Beginning of Android ADK with Arduino
2. Michael Margolis: Arduino Cook Book, O'Reilly, 2nd Edition

Reference Books:

1. Michael McRoberts: Beginning Arduino ,2nd Edition
2. Alan G. Smith: Introduction to Arduino

Web References:

1. https://www.tutorialspoint.com/arduino/[For Programming and Function Libraries]
2. http://www.instructables.com/id/Blink-Your-First-Arduino-Project/[LED Blink Project Step by Step]
3. https://diyhacking.com/esp8266-gpio/[Setting up Arduino Hands on]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Learn about microcontroller based system

C02 : Able to select appropriate family of microcontroller for different application.

C03 : Interface relevant hardware for given application.

C04 : Integrate hardware and software for embedded system

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	C02	C03	C04
1	Introduction to Arduino and Development Board	✓	✓		
2	Arduino Hardware and Software			✓	✓
3	Arduino Programming			✓	✓
4	GPIO Programming			✓	✓
5	Sensor and Actuators			✓	✓
6	Wireless Communications using Arduino			✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	1	3	2	2	1	1	1
CO2	2	3	2	1	3	2	2	1
CO3	1	1	3	3	3	2	2	3
CO4	1	2	1	3	3	3	1	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA608 Object Oriented System Analysis and Design

(Marks 100)

Contact Hours: 03

Pre- requisite: None.

Methodology & Pedagogy: This course focuses on providing hands-on experience in designing and developing large-scale software systems with emphasis on the use of analyzing real world application and techniques that enable large-scale software development. Students will generate concrete software engineering artifacts at all stages of the software life-cycle. Design principles and methods for large-scale software system development using object oriented software engineering.

Outline of the Course:

Sr No.	Practical	Description
Week 1	Concept of Software Development Life Cycle Assigning a problem definition and make students understand the problem domain	Students will be assign real world definition to understand
Week 2 Week 3	Gathering requirements and analyzing requirements	Students will gather the information for respective definition using various requirement gathering techniques
Week 4	Feasibility study of gathered requirements, Introduction to UML(Unified Modeling Language)	Students will create event table on basis of gathered requirement
Week 5 Week 6	Creating Use Case Diagram	Students will create use case diagram using event table which specifies functional requirement of the problem domain.

Week 7	Creating Activity Diagram	Students will create activity diagram which specifies flow of events for the problem domain.
Week 8 Week 9	Creating Class Diagram and Object Diagram	Students will analyze and identify the enities in form of classes on the basis of use case diagram and activity diagrams And classes will also be implemented using any object orientd programming language .
Week 10 Week 11	Creating Sequence and collaboration Diagrams	Students will draw sequence diagram and collobration diagram on basis of knowledge acquire from use case and class diagram which are base for interaction diagram, it will also include the implementation using any object oriented programming language.
Week 12	Review of sessions	Revision of all taken sessions.

Total Hours: 36

Core Books:

1. Timothy C. Lethbridge and Robert Laganière: Object-Oriented Software Engineering: Practical Software Development using UML and Java : Second Edition : McGraw-Hill Education: 2005
2. Bernd Bruegge : Object oriented software engineering :Second Edition, Pearson Education.
3. Roger Pressman: Software Engineering: Sixth Edition: Tata McGraw Hill.

Reference Books:

1. Grady Booch, James Rambaugh, Ivar Jacobson: The Unified Modeling Language User Guide, Addison Wesley.
2. Jacobson, Booch, Rumbaugh :The Unified Software Development Process : Pearson Education 1999.
3. Stephan R. Schach : Object oriented software engineering :Tata McGraw Hill.

Web References:

1. http://pl.cs.jhu.edu/oose/lectures
2. http://www.site.uottawa.ca/school/research/lloseng/supportMaterial/videos/
3. http://www.cs.uic.edu/~jbell/CourseNotes/OO_SoftwareEngineering
4. http://www.abssw.com/papers/UML_Overview.pdf

Course Outcomes: Upon successful completion of the course, students will:

C01 : Understand basic concepts of system development life cycle, overview of object orientation.

C02 : Gain knowledge about how the system will develop with its documentation using UML diagrams.

C03 : Learn about various types of UML behavioral diagrams such as Use case diagram, activity diagram, interaction diagram.

C04 : Understand UML structural diagram like Class Diagram, Object Diagram.

Course Outcomes Mapping:

Week Number	Topic Name	Course Outcomes			
		C01	C02	C03	C04
1-3	Basics of Information systems, SDLC , Overview of object oriented concepts	✓			
4	Basic concepts of UML	✓	✓		
5-6,10-11	Unified modeling language, Types of UML behavioral diagram	✓	✓	✓	
8-9	UML structural diagram such as Class and Object diagram.	✓	✓	✓	✓
12	Review of all sessions	✓	✓	✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	3	3	3	3
CO2	3	3	3	3	3	2	2	3
CO3	3	3	2	3	2	3	2	2
CO4	2	2	3	3	2	2	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS 124.01 C:
PROFESSIONAL COMMUNICATION (Sem-V)

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
V	HS124 A/B/C/D/E/F/G/H	Professional Communication	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction Professional Communication** • *Concept & Applications of Professional Communication* • *Rhetoric in Professional Communication* • *Importance of Ethos, Logos, and Pathos in Professional Communication*	**03**
2	**Cross-cultural Communication and Globalization** • *Basic Concepts: Culture, Globalization andCross-cultural Communication* • *Social and People Skills* • *Communicating with People of Different Cultures* • *Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them*	**08**
3	**Group Discussion and Personal Interviews** • *Cover Letters and Resume* • *Styles, Formats and Content of Cover Letters* • *Types of Resume* • *Concept and Rationale of Group Discussion* • *Skills and Aspects assessed in Group Discussion* • *Concept and Rationale of Personal Interview* • *Types of Personal Interview*	**10**

4	**Group Dynamics and Leadership** • *An Introduction to Group Dynamics and Leadership* • *Groups and their Structures* • *Roles and Functions of Members in Groups* • *Leading a Group* • *Types of Leadership/Leaders* • *Roles and Functions of a Leader* • *Characteristics of an effective Leader*	**05**
5	**Statement of Purpose (SOP)** • Concept and Rationale of Statement of Purpose • Statement of Purpose as a part of Selection Process • Types, Format and Nature of Statement of Purpose • Content and Process of Statement of Purpose	**04**
	Total	**30**

III. Pedagogy

Teaching will be facilitated by reading material, discussions, task-based learning, projects, assignments and interpersonal activities like group work, independent and collaborative study projects and presentations, etc.

IV. Evaluation

Internal Evaluation

Students' performance in the course will be evaluatedon a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment	02	05	10
2	Project	01	15	15
3			Attendance	05
			Total	30

External Evaluation

University Practical Examination will be for 70 marks to be conducted at the end of the semester. Details are:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical / Viva	01	70	70
			Total	70

Course Outcome (COs):

After completion of the course, the student would:

CO1 Gain basic conceptual understanding of communication skills in Professional settings

CO2 Develop awareness and competence in communicating across cultures in professional settings

CO3 Develop confidence and competence in speaking in formal interviews and to make formal presentations

CO4 Develop necessary soft skills to work collaboratively in a group and participate in a group Discussion for employment and in the workplace.

CO5 Develop team and group dynamics to work well in multi-disciplinary and cross-cultural work environment

CO6 Develop writing skills to prepare CV's,Resumes, Statement of Purpose and preparation of other formal documents.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	-	-	-	3	1	-
CO2	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	3	-	-
CO5	-	-	-	-	-	3	-	-
CO6	-	-	-	-	-	3	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books

- Koneru, A. *Professional Communication*, Tata McGrow Hill Education Private Limited
- Disanza, J.R. &Legge, N. *Business and Professional Communication,* Pearson Education
- Anandamurugan, A. *Placement Interviews – Skills for Success,* Tata McGrow Hill Education Private Limited
- Raman, M & Singh, P. *Business Communication,* Oxford University Press
- Adair, J. *Adair on Leadership*, CREST Publishing House

Teaching and Examination Scheme (B.Sc.(IT) Programme)

Effective from Year 2019-20

Semester-VI

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA609	Minor Project	-	12	12	12	-	-	-	50	50	200	300
CA610	Mobile Application Development	4	3	7	7	10	20	70	15	15	70	200
CA611-615	Elective - III	3	3	6	6	10	20	70	15	15	70	200
HS134 C	Contributor Personality Development	-	2	2	2	-		-	30		70	100
		07	20	27	27	200			600			800

Elective-III

1. CA611 Advanced Multimedia
2. CA612Open Source Framework - II
3. CA613 Scripting Framework - II
4. CA614 Data Mining & Analytics
5. CA615 Embedded Programming Using Raspberry Pi

CA609: Minor Project

(300 Marks)

Guidelines: Minor Project is in house project development. Every student is required to carry out Minor Project work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Minor Project work during the semester as per the schedule provided by the placement Coordinator.

Minor Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Minor Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of Minor Project to the institute. Each Student is required to make a copy of Minor Project in CD and submit along with Minor Project report.

Course Outcomes :

CO1 : Student will understand the implementation of concepts of SDLC and Software Engineering.

CO2 :The programming concepts they learn during their academics, it will be converted in to the actual implementations.

CO3: Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.

CO4: Students must understand the difference between a program and professional application/product/software.

CO5: Students will learn different categories of applications like Desktop application, Web applications, etc.

Evaluation: The project report shall normally be written in English in the specified format and shall be characterized by significant contribution to knowledge in the field. The Project report prepared according to approved guidelines and duly signed by the guide and the Head of the Department shall be submitted to the Head of the Institution. The evaluation scheme of Project is as under:

Course	Course Title	Teaching Scheme		Internal	End Semester Examination		Total
		Contact Hours	Credit	Continuous Evaluation	Report	Presentation & Viva	
CA609	Minor Project	12	12	100	100	100	300

The internal evaluation of project is done based on progress reports and internal presentations. The final evaluation of the project will be based on the project report submitted and a Viva-Voce Examination by a Board of Examiners.

If a candidate fails to submit the project report on or before the specified deadline, he/she is deemed to have failed in the Project Work and shall re-enroll for the same in a subsequent semester. If a candidate fails in the viva-voce examinations of Project work he/she shall resubmit the project report within specified duration decided by university. The resubmitted project will be evaluated during the subsequent academic session. A copy of the approved project report after the successful completion of viva examinations shall be kept in the library of the college / institution.

Web References:

1. http://techwhirl.com/writing-software-requirements-specifications/[For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.html [For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf [Requirement analysis guidelines]
4. http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf[Software Design Principles and Guidelines]
5. http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%20Testing%20-2050%20specific%20ways%20to%20improve%20your%20testing.pdf[ForEffective Software Testing]
6. http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template .pdf[For guidelines to prepare software project report]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	3	3	2	3
CO2	3	3	3	3	3	3	2	3
CO3	3	3	3	3	3	3	2	3
CO4	3	3	3	3	3	3	2	3
CO5	3	3	3	3	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA610: Mobile Application Development

(200 Marks)

Contact Hours: 07

Methodology & Pedagogy: During theory lectures, it describes various wireless technologies and basic features of Mobile Application Development. During Practical sessions, students will develop Mobile Application using JAVA programming language in Android. Student will also develop applications that handles data storage, documents sharing among applications and application based on Google maps.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Wireless Communication Fundamentals	06	36
2	Telecommunication Network	10	
3	Introduction and Basics of Android	07	
4	Understanding User Interface Design in Android	09	
5	Working with Adapters, Widgets, Alerts, Menus, Intents, Activities, Preferences	07	
6	Advanced Android Features	09	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit - I: Wireless Communication FundamentalsHours: 06
Introduction, Wireless transmission, Frequencies for radio transmission, Signals, Antennas, Signal Propagation, Multiplexing, Modulations, Spread spectrum, SDM, FDM, TDM, Cellular Wireless Networks

Unit - II: Telecommunication Network Hours: 10
Telecommunication systems Overview: GSM, GPRS, Satellite Networks, GSM, GPRS, CDMA

Unit - III: Introduction and Basics of Android Hours: 07
Overview of Android and Android API levels, Setting up development environment, Dalvik Virtual Machine and .apk file extension

Understanding of Fundamentals:

a. Basic Building blocks: Activities, Services, Broadcast Receivers & Content providers
b. UI Components: Views & notifications
c. Components for communication: Intents & Intent Filters

Application Structure: AndroidManifest.xml, Resources & R.java, Assets, uses-permission & uses-sdk, Activity/services/receiver declarations, Values: strings.xml, Layouts &Drawable Resources, Activities and Activity lifecycle, Toast, working with Emulator: Android Virtual Device, Deploying sample application on a real device, Logcat usage, Introduction to DDMS, File explorer

Unit - IV: Understanding User Interface in Android Hours: 09

Overview of UI design, Working with containers: LinerLayout, RelativeLayout, TableLayout, Scrollwork, FrameLayout. Basic Views: TextView, EditText, Button, ImageButton, CheckBox, ToggleButton, RadioButton, RadioGroup, ProgressBar, AutoComplete TextView, Picker Views: TimePicker and DatePicker.

Unit - V: Working with Adapters, Widgets, Alerts, Menus, Intents, Activities, Preferences Hours: 07 Adapters: ArrayAdapter, BaseAdapter, ListView and ListActivity, GridView, Gallery, AlertDialogs, Menus: Option menu, Context menu, Intents: Explicit Intents, Implicit Intents,Tabs and TabActivity ,Preferences: SharedPreferences, Preferences from xml

Unit - VI: Advanced Android Features Hours: 09

Working with SQLite Database, Overview of SQLiteOpenHelper class, Cursor, Broadcast Receivers: Understanding and implementing Broadcast Receiver, Services: Difference between Activity and Service, working with Android Services, Working with Web Service in Android, GoogleMap Integration, Using SD cards: Reading and writing data, Accessing Phone services (Call,SMS)

Core Books:

1. Jochen Schiller, “Mobile Communications”, PHI/Pearson Education, 2nd Edition, 2003.
2. William Stallings, “Wireless Communications and Networks”, PHI/Pearson Education, 2002.
3. Reto Meier: Professional Android 2 Application Development, Wrox publication 4.Mark L. Murphy: The Busy Coder’s Guide to Android Development

Reference Books:

1. Uwe Hansmann, LotharMerk, Martin S. Nicklons and Thomas Stober: Principles of Mobile Computing, Springer, New York, 2003.
2. Jonathan Simon: Head First Android Development, O’REILLY publication.
3. Mark L. Murphy: Beginning Android 2, APRESS publication.

Web References:

1. https://www.tutorialspoint.com/android/[Tutorial]
2. https://developer.android.com/training/index.html[Tutorial]
3. https://www.androidhive.info/[Practical Implementation Tutorial]
4. http://www.andrew.cmu.edu/course/95-702/slides/03_Android.pdf [Tutorials]
5. www.eli.sdsu.edu/courses/fall09/cs696/notes/index.html - United States [Control Tutorials]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Learn about wireless and telecommunication networks

C02 : Able to understand basic concepts and technique of Android App Development

C03 : Get knowledge about basic GUI development and Animation Effects

C04 : Learn about data management using Files, Shared Preferences

C05 : Able to work with database and web services

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Wireless Communication Fundamentals	✓				
2	Telecommunication Network	✓				
3	Introduction and Basics of Android		✓			
4	Understanding User Interface Design in Android		✓	✓		
5	Working with Adapters, Widgets, Alerts, Menus, Intents, Activities, Preferences			✓	✓	
6	Advanced Android Features			✓	✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	2	3	1	-	2
CO2	3	3	3	2	3	1	-	2
CO3	3	3	3	2	3	1	-	3
CO4	3	3	3	2	3	1	-	3
CO5	3	3	3	2	3	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA611: Advanced Multimedia

(200 Marks)

Contact Hours: 06

Pre-requisite: CA603 Fundamentals of Multimedia.

Methodology & Pedagogy: During theory sessions topics related to incorporate sound, video and animation in multimedia will be covered with suitable examples. During Practical sessions, students will be required to develop and produce advanced animations using appropriate tool.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	**Practical**
1	Working with Animation	**05**	**36**
2	Using Video in Multimedia	**07**	
3	Making Multimedia	**06**	
4	Designing for the World Wide Web	**06**	
5	Working with Web Animations	**06**	
6	Interactive Animations	**06**	

Total Hours (Theory): 36
Total Hours (Lab): 36
Total: 72

Detailed Syllabus:

Unit – I Working with Animation Hours:05

Introduction, The Power of Motion, Principles of Animation, Animation by Computer, Making Animations That Work

Unit – II Using Video in Multimedia Hours:07

Using Video, How Video Works and Is Displayed, Digital Video Containers, Obtaining Video Clips, Shooting and Editing Video.

Unit – III Making Multimedia Hours:06

The Stages of a Multimedia Project, What You Need: The Intangibles, Hardware, Software, Authoring Systems, Multimedia Team

Unit – IV Designing for the World Wide Web Hours:06

Developing for the Web, Text for the Web, Images for the Web, Sound for the Web, Animation for the Web, Video for the Web

Unit – V Working with Web Animations Hours:06

Understanding the Interface of Animation Tool - The Document Window, The Timeline, The Layer Controls, The Toolbar, The Panels

Using the Drawing and Color Tools - Drawing Tools Defined, Lines, Strokes, and Fills Defined, Drawing with the Pencil Tool, Using the Oval and Rectangle Tools, Using the Brush Tool, Deco brush, Modifying Strokes and Fills, Understanding the Drawing Models, The Merge and Object Drawing Models, Grouping Objects, Creating Gradients

Animate Your Art- The Timeline, Document Properties, Understanding Keyframes and Frames, What Is the Frame Rate? , Inserting and Deleting Frames , Copying and Reversing Frames, Testing and Publishing animated file,

Development of Animation- Frame-by-Frame Animation with Keyframes, What is Tweeing?, Understanding Basic Tweening, Using a Motion Guide , Animating Text , organizing layers, Layer Masking, Animating with the Blur Filter ,Animating with the Drop Shadow Filter

Unit – VI Interactive Animations Hours:06

Symbols and Templates - The Symbol and Instance Structure, Symbol Naming Convention, Creating Graphic Symbols, Creating Movie clip Symbol, Graphic symbol V/s Movie Clip Symbol, Creating Button Symbols, Creating Rollover Buttons with Text, Templates, Opening a Prebuilt Template, Customizing the Photo Album Template

Multimedia Scripting - Introduction to Scripting in Multimedia, Using Data Types, Variables, and Constants, Conditionals and Loops

Controlling Actions with Events- How Events Work, Mouse Events, Keyboard Events, Keeping Time with Timer Event

Core Books:

1. Principles of Multimedia by Ranjan Parekh. Tata McGraw-Hill
2. Multimedia: Making it work by Tay Vaughan - TMH, Eighth edition. 2006
3. Flash CS5-The Missing Manual by Chris Grover - O'Reilly publication

Reference Books:

1. Multimedia systems by John F. Koegal Buford-Pearson Education.
2. Multimedia Systems Design by Prabhat K. Andleigh and Kiran Thakrar-PHI publication 3.
   Adobe Flash Professional CS5 by Todd Perkins - Wiley Publication

Web References:

1. http://www.freeadobeflashtutorials.com/ - [Flash tutorial]

2. http://www.1stoptutorials.com/Flash-Beginners-Tutorial-Course.html - [Introduction to Flash]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Be able to identifies design challenges, examines the current and emerging technologies that support advanced multimedia systems.

C02: Understand and edit various file formats for audio, video and text media.

C03 : Able to understand the requirements for multimedia project and create business presentation and games.

C04 : Explore advanced web animation techniques.

C05 : Be able to Add interactivity to animations with buttons, menus, and scripting

Course Outcomes Mapping:

Unit No.	Content	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Working with Animation	✓				
2	Using Video in Multimedia	✓	✓			
3	Making Multimedia	✓		✓		
4	Designing for the World Wide Web	✓				✓
5	Working with Web Animations	✓			✓	
6	Interactive Animations	✓		✓		✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	3	2	2	2	2	-	2
CO2	2	3	2	2	2	2	-	2
CO3	2	3	2	2	2	1	-	2
CO4	2	3	2	2	2	1	-	2
CO5	2	3	2	2	2	1	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA612: Open Source Framework-II

(200 Marks)

Contact Hours: 06

Pre-requisite: CA604 Open Source Framework-I

Methodology & Pedagogy: During the theory lectures, concepts for MVC Framework, Blade template and different routing & filter methods will be learned. Also application creation using MVC & Blade template will be taught. The web applications used in the real world will be discussed with necessary examples. During the laboratory hours' students will implement the concepts that are learned during lectures.

Outline of the Course:

Unit. No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Laravel	05	36
2	Routing and Controllers	06	
3	Filters	05	
4	Blade Templates	06	
5	Database Connectivity	08	
6	State Management	06	

Total Hours (Theory): 36
Total Hours (Lab): 36
Total: 72

Detailed Syllabus:

Unit-I: Introduction to Laravel Hours: 05
Introduction of Laravel, Features of Laravel, composer, Architecture: request lifecycle, Service Container, Service Providers, Facades, Contracts, Directory Structure, Installation & Configuration.

Unit-II: Routing and Controllers Hours: 06
Route: Definition, Verbs, Handling, Parameters, Name routes, Groups, route model binding Middleware, Path Prefixes, Requests, Responses, Views, URL Generation, Validation, Errors & Logging.

Unit-III: Filters Hours: 05
Basic Filters, Multiple Filters, Filter Parameters, Filter Classes, Global filters, Default Filters, Pattern Filters.

Unit-IV: Blade Templates Hours: 06
Introduction to Blade Template, Template Inheritance, Components & Slots, Control Structures, File uploading, Send Email.

Unit-V: Database ConnectivityHours: 08
Introduction & Configuration of Database, connecting to a Database, Running Queries, Generating Query Results.

Unit-VI: State Management Hours: 06
Request: Retrieving the request URI, Retrieving Input, Cookie: Creating Cookie, Retrieving Cookie, Response: Basic Response, Attaching Headers, Attaching Cookies, Session: Creation and Retrieval.

Core Books:

1. Laravel Applica on Development Blueprints by ArdaKilicdagi Author , Halil Ibrahim Yilmaz (Author) [PACKT] Publishing
2. Code Bright- web application development with the laravel framework version 4 for beginners by Dayle Rees, Leanpub

Reference Books:

1.Laravel 4 Cookbook by Christopher Pitt, Leanpub

Web References:

1. https://laravel.com/docs/5.5/routing#form-method-spoofing[Form methods]
2. https://www.tutorialspoint.com/laravel/laravel_tutorial.pdf[Laravel Tutorial]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Be able to configure and install the Laravel framework.
C02 : Able to create MVC based application using the Laravel framework.
C03 : Understand Template integration and database communication in the web application development.
C04 : Able to incorporate different filtering techniques.
C05 : Learn various state management techniques.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Laravel		✓			
2	Routing and Controllers	✓	✓			
3	Filters		✓	✓	✓	
4	Blade Templates			✓		
5	Database Connectivity			✓		✓
6	State Management			✓		✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	2	2	2	-	-	2
CO2	2	3	2	2	2	-	-	2
CO3	2	3	2	2	2	-	-	2
CO4	2	3	2	2	2	-	-	2
CO5	3	3	2	2	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA613: Scripting Framework - II

(200 Marks)

Contact Hours: 06

Pre-requisite: CA605 Scripting Framework-I

Methodology & Pedagogy: During theory sessions topics related to common technologies and techniques used in the developing of web-based applications will be covered with suitable examples. During Practical sessions, students will be required to design and develop entire web sites using ROR.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Basics of Ruby Language	06	36
2	Standard types, Methods and Expression	07	
3	Arrays, Hashes and Enumerable	06	
4	Times and Dates, Exceptions, Catch and Throw	05	
5	Files, I/O and Database	06	
6	OOP and Dynamic features in Ruby	06	

Total Hours (Theory): 36
Total Hours (Lab): 36
Total: 72

Detailed Syllabus:

Unit – I: Basics of Ruby Language Hours: 06

What is Ruby; Ruby download and installation, An Introduction to Object Orientation, Basic Ruby Syntax and Semantics OOP in Ruby, Dynamic Aspects of Ruby.

Unit – II: Standard types, Methods and Expression Hours: 07

Numbers, Strings, Ranges, Regular Expressions, defining a method, Calling a method, Operator expressions, Miscellaneous expressions, Assignment, Conditional execution, Case expression, Loops, variable Scope, Loops and Blocks.

Unit – III: Working with Arrays, Hashes and Enumerable Hours: 06

Comparing arrays, Sorting array, selecting from an array by criteria, using multidimensional arrays, Finding, transforming, removing, Concatenating and reversing in arrays and hashes with key value pairs. Searching and selecting, Counting and comparing, iterating, Extracting and converting in Enumerable

Unit – IV: Times and Dates, Exceptions, Catch and Throw Hours: 05

Determining and finding time and dates, conversion, parsing and formatting of date and time. Exception class, Handling and Raising exceptions, Catch and Throw.

Unit – V: Files, I/O and Database Hours: 06

What is an IO object, Opening and closing files, Reading and writing files, Connecting to external data stores like MySQL, ORM and SQLite.

Unit – VI: OOP and Dynamic features in Ruby Hours: 06

Revision of OPP concepts, Learning some advance techniques and dynamic features and Program introspection.

Core books:

1. The ruby way, Third edition by Hal Fulton with Andre Arko, Addison Wesley.
2. Programming Ruby, The Pragmatic Programmers' Guide, Second Edition by Dave Thomas with Chad Fowler and Andy Hunt, The Pragmatic Bookshelf Raleigh, North Carolina Dallas, Texas
3. The Well-Grounded Rubyist by DAVID A. BLACK, MANNING Publications Co, Greenwich.

Reference Books:

1. Beginning Ruby by Peter Cooper, Apress
2. Learn to Program by Chris Pine, The Pragmatic Bookshelf

Web Reference:

1. https://codemy.com/learning-ruby-rails-web-development/[For Video tutorial on Ruby]
2. https://www.javatpoint.com/ruby-on-rails-tutorial[For basics of Ruby]
3. http://www.xyzpub.com/en/ruby-on-rails/3.2/ruby-array-und-hash.html[For examples of array and hash]
4. http://guides.rubyonrails.org/v2.3/getting_started.html[For database using Ruby]
5. https://medium.com/rails-ember-beyond/error-handling-in-rails-the-modular-way-9afcddd2fe1b[For exception handling using Ruby]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Understand the basic concepts of Scripting Framework using Ruby Programming Language.

C02 : Be able to design Simple Web Application using Ruby Programming Language.

C03 : Be able to understand concepts of File,I/O and databases using Ruby.

C04: Be able to understand dynamic features of Ruby for real time complex system.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	C02	C03	C04
1	Basics of Ruby Language	✓	✓		
2	Standard types, Methods and Expression	✓	✓		
3	Arrays, Hashes and Enumerable	✓	✓		
4	Times and Dates, Exceptions, Catch and Throw	✓	✓		
5	Files, I/O and Database			✓	
6	OOP and Dynamic features in Ruby				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	2	2	2	2	-	-	2
CO2	1	3	2	2	2	-	-	2
CO3	1	2	2	2	2	-	-	2
CO4	2	3	2	2	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA614: Data Mining & Analytics

(Marks 200)

Contact Hours: 06

Pre-requisite: CA606 Advanced Data Science

Methodology & Pedagogy:

During theory lectures Data Mining studies algorithms and computational paradigms that allow computers to find patterns and regularities in databases, perform prediction and forecasting, and generally improve their performance through interaction with data will be covered. During practical session the students will use recent Data Mining software, for implementing the algorithms.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Introduction to Data Mining and Warehousing	06	36
2	Data Preprocessing	05	
3	Association Rule Mining	07	
4	Classification and Prediction	06	
5	Clustering Techniques	06	
6	Advanced techniques and applications	06	

Total Hours (Theory): 36
Total Hours(Practical):36
Total: 72

Detailed Syllabus:

Unit – I: Introduction to Data Mining and Warehousing Hours: 06
Data mining concept, Data Mining Goals, Stages of the Data Mining Process,Data Mining Techniques, Knowledge Representation Methods, Data Warehouse and OLAP, Data Warehouse and DBMS, Multidimensional data model, OLA Poperations, Applications, Example: weather data, loan data, etc.

Unit – II: Data Preprocessing Hours:05
Data cleaning, Data transformation, Data reduction, Discretization and generating concept hierarchies

Unit – III: Association Rule Mining Hours:07
Motivation and terminology, Basic idea of item sets, Generating item sets and rules efficiently Correlation

Analysis

Unit – IV: Classification and Prediction Hours:06
Basic learning/mining tasks, inferring rudimentary rules: 1R algorithm, Decision trees, Covering rules The prediction task, Statistical (Bayesian) classification, Bayesian networks, Instance-based methods (nearest neighbor), Linear models

Unit – V: Clustering Techniques Hours:06
Basic issues in clustering, first conceptual clustering system: Cluster/2, Partitioning methods: kmeans, expectation maximization (EM), Hierarchical methods: distance-based agglomerative and divisible clustering, Conceptual clustering: Cobweb

Unit – VI: Advanced techniques and applications Hours:06
Text mining: extracting attributes (keywords), structural approaches (parsing, soft parsing).
Bayesian approach to classifying text
Web mining: classifying web pages, extracting knowledge from the web Data Mining software and applications

Core Books:

1. Jiawei Han &MichelineKamber: Data Mining: Concepts & Techniques, 2nd Edition, Morgan Kaufmann Publishers.

Reference Books:

1. PaulrajPonniah: Data Warehousing Fundamentals: A Comprehensive Guide for IT Professionals, Wiley-India.
2. Daniel T. Larose: Data Mining Methods & Models, Wiley-India.
3. VikramPudi& P. Radhakrishnan: Data Mining, Oxford University Press.
4. Alex Berson& Stephen J. Smith: Data Warehousing, Data Mining & OLAP, Tata McGraw-Hill.

Web References:

1. https://www.sas.com/en_us/offers/sem/data-mining-from-a-z-104937.html[Data Mining Notes]
2. https://docs.microsoft.com/en-us/sql/analysis-services/data-mining-tutorialsanalysisservices[Data Mining notes on Analysis Services]
3. http://www.tutorialspoint.com/data_mining/[Data Mining Tutorial]
4. http://www.zentut.com/data-mining/ [Data Mining Tutorial]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Come to know the basic concepts of data mining and warehousing

C02 : Understand various data pre-processing techniques

C03 : Learn different techniques to discover the association rules.

C04 : Gain in-depth understanding of various techniques for classification, prediction, and clustering

C05 : learn advanced technique for text mining and web mining

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to Data Mining and Warehousing	✓				
2	Data Preprocessing		✓			
3	Association Rule Mining			✓		
4	Classification and Prediction				✓	
5	Clustering Techniques				✓	
6	Advanced techniques and applications					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	-	2	-	1	1	2
CO2	3	-	1	-	2	1	-	-
CO3	-	-	-	3	1	-	1	-
CO4	2	2	3	-	-	2	-	1
CO5	3	1	2	3	1	-	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA615: Embedded Programming using Raspberry Pi

(Marks 200)

Contact Hours: 06

Pre- requisite: CA607 Embedded Programming using Arduino

Methodology & Pedagogy: During theory lectures, students will primarily emphasis on Raspberry Pi set up and its boot configuration. They will have good hands on basics of Python as well as Raspberry Pi and how both can be integrated. In the practical sessions, students will begin with basic programming using Pi and eventually learn about its architecture. In addition, basic input output will be carried out using button and LED with GPIO. Various practices will also be imparted using Graphical User Interface.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	**Practical**
1	Introduction to Raspberry Pi	06	36
2	Introduction to Python Programming Language	08	
3	Raspberry pi with Python	06	
4	Raspberry pi Programming	04	
5	Basic Input and Output	06	
6	Graphical User Interfaces	06	

Total Hours (Theory): 36
Total Hours (Lab): 36
Total: 72

Detailed Syllabus:

Unit - I: Introduction to Raspberry Pi Hours: 06

Introduction to Raspberry Pi, Why Raspberry pi, Features of Raspberry pi, Basic set up and first boot configuration, Different uses of Raspberry pi, Different Versions of Raspberry pi, Vision of Raspberry pi, Different components of the Board.

Unit - II: Introduction to Python Programming Language Hours: 08

Variables and Numbers, Looping Structures, Conditional Statements, Lists, Tuples and Dictionaries, Type Conversions, Function declaration, calling functions and passing values, Function Returning values. Exiting from functions, Reading/Writing to File.

Unit - III: Raspberry Pi with Python Hours: 06

Why Raspberry Pi and Python, what you can do with Raspberry Pi and Python, writing a first Python application, GPIO programming using python.

Unit - IV: Raspberry Pi Programming Hours: 04
Basic Programming of the Pi, Hello World program, Access the World Wide Web, Play audio, Control Peripherals with a Pi (Raspberry Pi chip), Architecture of Raspbian OS, Raspberry Pi Cabling/Setup/Startup

Unit - V: Basic Input and Output and GPIO Hours: 06
Using Inputs and Outputs: Digital Output - Lighting Up an LED, Digital Input - Reading a Button, Testing GPIO in Python, Pin numbering Formats of GPIO, The LED Interfacing, The First Button Interface with Raspberry Pi, General information on other pins and their functionality, blinking an LED, Reading a Button.

Unit – VI: **Graphical User InterfacesHours: 06**
Tkinter, Hello World program, Temperature Converter, Other GUI Widgets, Dialogs Menus, The Canvas.

Core Books:

1. Matt Richardson and Shawn Wallace: Getting Started with Raspberry Pi,3rd Edition
2. Dr. Simon Monk: Programming the Raspberry Pi: Getting Started with Python, McGraw Hill Publication

Reference Books:

1. Wolfram Donat: Learn Raspberry Pi Programming with Python
2. M. Richardson, S. Wallace: Getting Started with Raspberry Pi, O'Reilly, 2012

Web References:

1. http://opensourceforu.com/2016/10/programming-raspberry-pi-with-python/[Raspberry Pi andpythonTogether]
2. https://www.raspberrypi.org/help/quick-start-guide/[Setting Up Raspberry Pi]
3. https://www.youtube.com/watch?v=UUOCh0Cbty8[Raspberry Pi - How to start programming with Python]
4. https://www.raspberrypi.org/documentation/usage/python/[Step by Step Programming]

Course Outcomes: Upon successful completion of the course, the students will :

Embedded/IoT applications using Python and Rpi

C01 :	Know about arm microprocessor and different types of Raspberry Pi
C02 :	Learn about of Python Programming Language and its variant
C03 :	Develop application programs in Python
C04 :	Implement applications on Raspberry Pi using python
C05 :	Develop and Implement Embedded/IoT applications using Python and Rpi

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to Raspberry Pi	✓				
2	Introduction to Python Programming Language		✓	✓		
3	Raspberry pi with Python				✓	
4	Raspberry pi Programming				✓	
5	Basic Input and Output				✓	✓
6	Graphical User Interfaces				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	-	-	2	-	-	3	-
CO2	3	2	-	-	-	2	-	3
CO3	-	-	3	-	2	-	2	-
CO4	-	-	1	-	2	2	-	2
CO5	-	-	-	2	-	1	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS134 C: CONTRIBUTOR PERSONALITY DEVELOPMENT (Sem- VI)

Credits and Schemes:

Sem.	Course Code	Course Name	Credits	Teaching Scheme Contact Hours/Week	Evaluation Scheme				Total
					Theory		Practical		
					Internal	External	Internal	External	
VI	HS134 A/B/C/D/E/F/G/H	CONTRIBUTOR PERSONALITY DEVELOPMENT	02	02	--	--	30	70	100

II. **Course Outline**

Module No.	Title/Topic	Classroom Contact Hours
1	**Concept of Personality:** • Meaning of Personality • Types of Personality • Factors contributing to Personality • Personality Traits	06
2	**Soft Skills and Personality Development:** • Critical, Creative and Positive Thinking • Leadership, Assertiveness and Negotiation Skills • Self-Management • People's Skills • Building Relationship Skills • Being a Team Player	08
3	**Developing Contributor Personality – Part I** • Concept of Contributor • Characteristics of a Contributor • The Contributor's Vision of Success & Career • The Scope of Contribution • Embarking on the Journey to Contributor-ship	06

4	**Developing Contributor Personality – Part II** • Focus on values • Engage deeply • Think in enlightened self-interest • Practice imaginative sympathy • Demonstrate trust behavior • Developing a sense of duty and morality	06
5	**Contemporary Issues in CPD** • Contemporary Practices & Trends in Contributor Personality Development • Case Study & Presentations	04
	Total	**30**

III. **Instruction Methods and Pedagogy**

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, critical reading, group work, independent and collaborative research, presentations etc.

IV. **Evaluation:**

The students will be evaluated continuously in the form of internal as well as external evaluation. It is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
3	Assignment / Project Work / Term Work / Quiz	5	5	25
4	Attendance and Class-room Participation			05
			Total	30

External Evaluation

The University Practical examination will be of 70 marks and will test the contributory personality aspects and their applications by carrying out practical assessment. The examination will avoid, as far as possible, evaluation on the basis of grammatical errors. Instead, it will focus on applications.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical Exam / Viva	01	70	70
			Total	70

Course Outcome (COs):

After completion of the course, the student would :

CO1 Develop conceptual understanding of the term Personality and understand one's own personality Traits

CO2 Develop soft skills for holistic personality development for career and in personal life

CO3 Develop assertiveness,self management and negotiation skills to navigate the personal and professional environment successfully

CO4 Develop conceptual clarity of the term ' Contributor Personality' and develop understanding of the traits of a contributor

CO5 Develop the qualities of a contributor such as trust, strong work ethic, responsibility and accountabililty at workplace

CO6 Develop relevant skills required to be a contributor at workplace and achieve success in life

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	3	-	-
CO3	-	-	-	-	-	3	-	-
CO4	-	-	-	-	-	3	2	-
CO5	-	-	-	-	-	3	-	-
CO6	-	-	-	-	-	3	-	3

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Reference Books / Reading

- Contributor Personality Program Workbook (Volume 1,2),
- Contributor Personality Program Active Guide, Illumine Knowledge Pvt. Ltd.